Linear Algebra and Its Applications

Chapter 6

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.1 Inner product, length, and orthogonality

For real columns u, v of equal length, $u \cdot v = u^T v$ is the sum of corresponding coordinate products. The norm is $\|v\| = \sqrt{v \cdot v}$, and distance is $\|u - v\|$. Two vectors are orthogonal precisely when their inner product is zero. Normalize a nonzero vector by dividing by its length, not by its squared length. The zero vector has no unit vector in its direction.

For a subspace W, its orthogonal complement $W^\perp$ contains the vectors orthogonal to *every* vector of W. Testing a spanning family suffices because the inner product distributes over linear combinations.

Practice problem 6.1.P1: A projection coefficient and its scaled vector

Compute $a \cdot b = (-2)(-3) + 1(1) = 7$ and $a \cdot a = (-2)^2 + 1^2 = 5$. Thus

$$\frac{a \cdot b}{a \cdot a} = \frac{7}{5}, \qquad \left(\frac{a \cdot b}{a \cdot a}\right) a = \begin{bmatrix} -14/5 \\ 7/5 \end{bmatrix}.$$

The first answer is a scalar, while the second is a vector.

Practice problem 6.1.P2: Normalize the three-coordinate vector

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{4}{3})^2 + (-1)^2 + (\frac{2}{3})^2} = \frac{\sqrt{29}}{3}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{4\sqrt{29}}{29} \\ -\frac{3\sqrt{29}}{29} \\ \frac{2\sqrt{29}}{29} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Practice problem 6.1.P3: Check orthogonality directly

The inner product is $5(4/3) + 6(-1) + (-1)(2/3) = 20/3 - 6 - 2/3 = 0$. Therefore d is orthogonal to c; checking the zero inner product is exactly the definition needed.

Practice problem 6.1.P4: Orthogonality survives normalization

The unit vector u from Practice 2 is $c/\|c\|$. By linearity in the second argument, $d \cdot u = (1/\|c\|)(d \cdot c) = 0$ using Practice 3.

Conclusion. Multiplying one of two orthogonal vectors by a scalar does not destroy orthogonality.

Exercise 6.1.1: Inner-product and norm calculations

Compute the two dot products from corresponding coordinates.

$$u \cdot u = (-1)(-1) + (2)(2) = 5,$$
$$v \cdot u = (4)(-1) + (6)(2) = 8.$$

Conclusion. The requested ratio is $(v \cdot u)/(u \cdot u) = \frac{8}{5}$.

Exercise 6.1.2: Inner-product and norm calculations

Compute the two dot products from corresponding coordinates.

$$w \cdot w = (3)(3) + (-1)(-1) + (-5)(-5) = 35,$$
$$x \cdot w = (6)(3) + (-2)(-1) + (3)(-5) = 5.$$

Conclusion. The requested ratio is $(x \cdot w)/(w \cdot w) = \frac{1}{7}$.

Exercise 6.1.3: Inner-product and norm calculations

The squared length is $w \cdot w = 35$. Multiply every coordinate by its reciprocal:

$$\frac{1}{w \cdot w} w = \frac{1}{35} \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} \frac{3}{35} \\ -\frac{1}{35} \\ -\frac{1}{7} \end{bmatrix}.$$

This is not generally a unit vector: normalization would divide by the square root of this denominator.

Exercise 6.1.4: Inner-product and norm calculations

The squared length is $u \cdot u = 5$. Multiply every coordinate by its reciprocal:

$$\frac{1}{u \cdot u} u = \frac{1}{5} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{5} \\ \frac{2}{5} \end{bmatrix}.$$

This is not generally a unit vector: normalization would divide by the square root of this denominator.

Exercise 6.1.5: Inner-product and norm calculations

The numerator is $u \cdot v = 8$ and the denominator is $v \cdot v = 52$. Therefore

$$\left(\frac{u \cdot v}{v \cdot v}\right) v = \frac{2}{13} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{8}{13} \\ \frac{12}{13} \end{bmatrix}.$$

The denominator is nonzero because the vector being projected onto is nonzero.

Exercise 6.1.6: Inner-product and norm calculations

The numerator is $w \cdot x = 5$ and the denominator is $x \cdot x = 49$. Therefore

$$\left(\frac{w \cdot x}{x \cdot x}\right) x = \frac{5}{49} \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{30}{49} \\ -\frac{10}{49} \\ \frac{15}{49} \end{bmatrix}.$$

The denominator is nonzero because the vector being projected onto is nonzero.

Exercise 6.1.7: Inner-product and norm calculations

By the norm definition,

$$\|w\| = \sqrt{(3)^2 + (-1)^2 + (-5)^2} = \sqrt{35} = \sqrt{35}.$$

The norm is the nonnegative square root, so there is no $\pm$ choice.

Exercise 6.1.8: Inner-product and norm calculations

By the norm definition,

$$\|x\| = \sqrt{(6)^2 + (-2)^2 + (3)^2} = \sqrt{49} = 7.$$

The norm is the nonnegative square root, so there is no $\pm$ choice.

Exercise 6.1.9: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(-30)^2 + (40)^2} = 50.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.10: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(-6)^2 + (4)^2 + (-3)^2} = \sqrt{61}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} -\frac{6\sqrt{61}}{61} \\ \frac{4\sqrt{61}}{61} \\ -\frac{3\sqrt{61}}{61} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.11: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{7}{4})^2 + (\frac{1}{2})^2 + (1)^2} = \frac{\sqrt{69}}{4}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{7\sqrt{69}}{69} \\ \frac{2\sqrt{69}}{69} \\ \frac{4\sqrt{69}}{69} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.12: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{8}{3})^2 + (2)^2} = \frac{10}{3}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.13: Inner-product and norm calculations

Subtract the point coordinates first:

$$a - b = \begin{bmatrix} 10 \\ -3 \end{bmatrix} - \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 11 \\ 2 \end{bmatrix}.$$

Conclusion. The distance is $\|a - b\| = \sqrt{125} = 5\sqrt{5}$. Reversing the subtraction gives the same length.

Exercise 6.1.14: Inner-product and norm calculations

Subtract the point coordinates first:

$$a - b = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix} - \begin{bmatrix} -4 \\ -1 \\ 8 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ -6 \end{bmatrix}.$$

Conclusion. The distance is $\|a - b\| = \sqrt{68} = 2\sqrt{17}$. Reversing the subtraction gives the same length.

Exercise 6.1.15: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (8)(-2) + (-5)(-3) = -1.$$

Conclusion. The pair is not orthogonal because the product is nonzero.

Exercise 6.1.16: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (12)(2) + (3)(-3) + (-5)(3) = 0.$$

Conclusion. The pair is orthogonal because the product is zero.

Exercise 6.1.17: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (3)(-4) + (2)(1) + (-5)(-2) + (0)(6) = 0.$$

Conclusion. The pair is orthogonal because the product is zero.

Exercise 6.1.18: Inner-product and norm calculations

Compute the defining dot product:

$$a \cdot b = (-3)(1) + (7)(-8) + (4)(15) + (0)(-7) = 1.$$

Conclusion. The pair is not orthogonal because the product is nonzero.

Exercise 6.1.19: Inner-product and norm calculations

(a) True. It is the squared-norm identity $\|v\|^2 = v \cdot v$.

(b) True. Scalar distributivity gives $u \cdot (cv) = c(u \cdot v)$.

(c) True. Squaring the equal distances and expanding gives $\|u\|^2 - 2u \cdot v + \|v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. Hence $4u \cdot v = 0$, so the vectors are orthogonal.

(d) False. Take $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Its column space contains $(1, 0)^T$ and its null space contains $(-1, 1)^T$; their dot product is -1, not zero. The correct general pairing is *row space* with null space.

(e) True. For $w = \sum_j c_j v_j$ we have $x \cdot w = \sum_j c_j (x \cdot v_j) = 0$; thus $x \in W^\perp$.

Exercise 6.1.20: Inner-product and norm calculations

(a) True. Symmetry gives $u \cdot v = v \cdot u$, so their difference is zero.

(b) False for negative scalars. For $c = -1$ and $v \neq 0$, $\|cv\| = \|v\|$ but $c\|v\| = -\|v\|$. The correct rule uses $|c|$.

(c) True. Being orthogonal to every member of W is the definition of membership in $W^\perp$.

(d) True. Expand $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. The given equality makes $2u \cdot v = 0$.

(e) True. $Ax = 0$ means every row has dot product zero with x. The same holds for every linear combination of rows, hence all of Row A.

Exercise 6.1.21: Inner-product and norm calculations

(b) Distribute using the transpose and matrix multiplication rules.

$$(u + v) \cdot w = (u + v)^T w = (u^T + v^T)w = u^T w + v^T w = u \cdot w + v \cdot w.$$

Here we used $(u + v)^T = u^T + v^T$ and distribution of matrix products across addition, both from Chapter 2.

(c) Move a scalar through the products.

The transpose rule $(cu)^T = cu^T$ and scalar associativity of products give

$$(cu) \cdot v = (cu)^T v = c(u^T v) = c(u \cdot v) = u^T (cv) = u \cdot (cv).$$

Each expression is a 1×1 product, identified with its real scalar value.

Exercise 6.1.22: Inner-product and norm calculations

We have $u \cdot u = u_1^2 + u_2^2 + u_3^2$. Every real square is nonnegative, so the sum is nonnegative. If the sum is zero, none of the nonnegative summands can be positive, so $u_i^2 = 0$ for every i. This forces all three entries to be zero. Conversely the zero vector has zero inner product with itself.

Conclusion. $u \cdot u \geq 0$, with equality if and only if $u = 0$.

Exercise 6.1.23: Inner-product and norm calculations

Compute directly rather than invoking the Pythagorean theorem:

$$u \cdot v = (2)(-7) + (-5)(-4) + (-1)(6) = 0,$$

$$\|u\|^2 = 30, \qquad \|v\|^2 = 101, \qquad u + v = \begin{bmatrix} -5 \\ -9 \\ 5 \end{bmatrix}.$$

$$\|u + v\|^2 = (-5)^2 + (-9)^2 + (5)^2 = 131 = 30 + 101.$$

The dot product is zero, so this example agrees with the orthogonal-sum identity, which has now been verified by arithmetic.

Exercise 6.1.24: Inner-product and norm calculations

Use bilinearity and symmetry to expand both squared norms:

$$\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2,$$

$$\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2.$$

Adding cancels the two cross terms and gives $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$.

Conclusion. The parallelogram law holds for all $u, v \in \mathbb{R}^n$; orthogonality is not assumed.

Exercise 6.1.25: Inner-product and norm calculations

Orthogonality to $(a, b)^T$ is the single homogeneous equation $ax + by = 0$. If $a = b = 0$, every (x, y) works, so $H = \mathbb{R}^2$. If $(a, b) \neq (0, 0)$, the nonzero vector $(-b, a)^T$ is orthogonal to $(a, b)^T$ and its multiples solve the equation. Conversely, if $a \neq 0$ solve $x = -(b/a)y$; if $a = 0$ then $b \neq 0$ and the equation gives $y = 0$. Either way every solution is a multiple of $(-b, a)^T$.

Conclusion. For $v \neq 0$, H is the line through the origin perpendicular to v; for $v = 0$, it is the entire plane.

Exercise 6.1.26: Inner-product and norm calculations

We can write $W = \text{Nul}[5\ -6\ 7]$. The Chapter 4 theorem that a null space is a subspace therefore applies. Explicitly its equation is $5x_1 - 6x_2 + 7x_3 = 0$, one nonzero homogeneous linear constraint. Solving for x_1 leaves two freely chosen coordinates, so the geometric set is a plane through the origin. The normal direction is u because $u \cdot x = 0$ for every vector in the plane.

Conclusion. W is the plane perpendicular to $(5, -6, 7)^T$ through zero.

Exercise 6.1.27: Inner-product and norm calculations

By distributivity, $y \cdot (u + v) = y \cdot u + y \cdot v = 0 + 0 = 0$. The two zero values follow from the two given orthogonality assumptions.

Conclusion. y is orthogonal to $u + v$.

Exercise 6.1.28: Inner-product and norm calculations

Take any $w \in \text{Span}\{u, v\}$. There are real scalars c_1, c_2 with $w = c_1u + c_2v$. Then $y \cdot w = c_1(y \cdot u) + c_2(y \cdot v) = 0$. Because w was arbitrary, the claim holds for every vector in the span, not merely for its two generators.

Conclusion. $y \in (\text{Span}\{u, v\})^{\perp}$.

Exercise 6.1.29: Inner-product and norm calculations

An arbitrary $w \in W$ can be written $w = \sum_{j=1}^{p} c_j v_j$. Apply distributivity repeatedly:

$$x \cdot w = \sum_{j=1}^{p} c_j (x \cdot v_j) = \sum_{j=1}^{p} c_j 0 = 0.$$

Thus every w in W is orthogonal to x.

Conclusion. Testing orthogonality against a spanning family tests all of W.

Exercise 6.1.30: Inner-product and norm calculations

(a) Closure under scalar multiplication.

Let $z \in W^{\perp}$, $c \in \mathbb{R}$, and let u be any vector of W. Then $(cz) \cdot u = c(z \cdot u) = c0 = 0$. Since this holds for every $u \in W$, $cz \in W^{\perp}$.

(b) Closure under addition.

For $z_1, z_2 \in W^{\perp}$ and any $u \in W$, $(z_1 + z_2) \cdot u = z_1 \cdot u + z_2 \cdot u = 0 + 0 = 0$. Hence $z_1 + z_2 \in W^{\perp}$.

(c) Include zero and apply the subspace test.

The zero vector has zero dot product with every $u \in W$, so it lies in $W^{\perp}$. Together with (a) and (b), these are all three required subspace properties.

Conclusion. $W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Exercise 6.1.31: Inner-product and norm calculations

If $x \in W$ and $x \in W^{\perp}$, then the latter membership makes x orthogonal to every member of W, including itself. Thus $x \cdot x = \|x\|^2 = 0$, which forces $x = 0$ by positive definiteness.

Conclusion. $W \cap W^{\perp} = \{0\}$.

Exercise 6.1.32: Inner-product and norm calculations

(a) Check all column lengths and mutual dot products.

The single exact calculation $A^T A = I_4$ states that every column has squared length 1 and all six off-diagonal pairwise products are zero. Thus the four lengths are $1, 1, 1, 1$ and the six requested dot products are 0.

(b)–(d) Perform three reproducible random experiments.

Using integer draws from $[-3, 3]$ with Python random seed 61032 gives the following three nonzero pairs. Listed equalities compare the original and transformed quantities.

$$u = \begin{bmatrix} 2 \\ 2 \\ 0 \\ -3 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \end{bmatrix};$$

$$\|u\| = \|Au\| = \sqrt{17}, \quad \|v\| = \|Av\| = \sqrt{6},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = -\frac{\sqrt{102}}{17}.$$

$$u = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 2 \end{bmatrix}, \quad v = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \end{bmatrix};$$

$$\|u\| = \|Au\| = \sqrt{14}, \quad \|v\| = \|Av\| = \sqrt{14},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = \frac{5}{14}.$$

$$u = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \quad v = \begin{bmatrix} 0 \\ 2 \\ -1 \\ -1 \end{bmatrix};$$

$$\|u\| = \|Au\| = 2, \quad \|v\| = \|Av\| = \sqrt{6},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = \frac{\sqrt{6}}{6}.$$

Prove the observed behavior for every pair.

For arbitrary u, v, $(Au)^T(Av) = u^T A^T Av = u^T v$. Taking $v = u$ proves norm preservation, and dividing by nonzero norms proves angle preservation for nonzero vectors.

Conclusion. A preserves lengths and angles; the numerical experiments are instances of this identity.

Exercise 6.1.33: Inner-product and norm calculations

Compute two actual random examples with one fixed nonzero direction.

Using seed 61033 and integer draws from $[-3, 3]$, take $v = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, with $v \cdot v = 5$. Each scalar below multiplies this same explicitly given column.

$$x = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 1 \end{bmatrix},$$

$$T(x) = (\frac{8}{5})v, \quad T(y) = (-\frac{3}{5})v,$$
$$T(x+y) = (1)v, \quad T(10x) = (16)v.$$
$$x = \begin{bmatrix} -2 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \quad y = \begin{bmatrix} -1 \\ 2 \\ 3 \\ -1 \end{bmatrix},$$
$$T(x) = (-\frac{1}{5})v, \quad T(y) = (-\frac{3}{5})v,$$
$$T(x+y) = (-\frac{4}{5})v, \quad T(10x) = (-2)v.$$

Verify the conjectured linearity algebraically.

For scalars a, b and any inputs x, y, distributivity gives

$$T(ax+by) = \frac{(ax+by)\cdot v}{v\cdot v}v = aT(x) + bT(y).$$

The fixed denominator is nonzero because $v \neq 0$.

Conclusion. Projection onto a fixed nonzero direction is linear; the displayed experiments satisfy additivity and scalar homogeneity exactly.

Exercise 6.1.34: Inner-product and norm calculations

Read the free variables from an exact row reduction.

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 5 & 0 & -\frac{1}{3} \\ 0 & 1 & 1 & 0 & -\frac{4}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are $1, 2, 4$, leaving free coordinates $3, 5$. A null-space basis, with the second vector scaled by 3 to avoid fractions, and a nonzero-row basis for the row space give

$$N = \begin{bmatrix} -5 & 1 \\ -1 & 4 \\ 1 & 0 \\ 0 & -1 \\ 0 & 3 \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 0 & 15 & 0 & -1 \\ 0 & 3 & 3 & 0 & -4 \\ 0 & 0 & 0 & 3 & 1 \end{bmatrix}.$$

The two columns of N have independent free-coordinate choices, and the three rows of R are scaled nonzero rows of RREF, so they are bases of the required spaces.

Multiply the two basis matrices.

$$RN = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Each entry is the dot product of one row-space basis vector with one null-space basis vector. All are zero, illustrating $(\text{Row } A)^{\perp} = \text{Nul } A$ from Theorem 3. The additional check $AN = 0$ verifies the null-space computation directly.

6.2 Orthogonal sets

An orthogonal set has pairwise zero inner products; it may include zero. An orthonormal set is orthogonal and has every length equal to 1, so its vectors are necessarily nonzero. For nonzero orthogonal columns u_j, a vector in their span has unique weights $c_j = (x \cdot u_j)/(u_j \cdot u_j)$.

For a line $L = \text{Span}\{u\}$, $u \neq 0$, the orthogonal projection of y is $\widehat{y} = ((y \cdot u)/(u \cdot u))u$. Its residual $y - \widehat{y}$ is perpendicular to the line; the distance to the line is the length of that residual. A rectangular matrix can have orthonormal columns, but the term *orthogonal matrix* is reserved here for a square matrix with $U^{-1} = U^T$.

Practice problem 6.2.P1: Two orthonormal vectors form a plane basis

Their dot product is $(-1/\sqrt{5})(2/\sqrt{5}) + (2/\sqrt{5})(1/\sqrt{5}) = 0$. Their squared lengths are $(1+4)/5 = 1$ and $(4+1)/5 = 1$. Thus they are orthonormal and in particular independent. Two independent vectors in $\mathbb{R}^2$ form a basis.

Conclusion. The proposed pair is an orthonormal basis of $\mathbb{R}^2$.

Practice problem 6.2.P2: The same projection using a different line generator

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 5 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{20}{5} = 4.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The projection is $(8, 4)^T$, the same point obtained with any nonzero generator of the same line.

Practice problem 6.2.P3: A rectangular isometry preserves inner products

The data from Example 6 are

$$U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}, \quad x = (\sqrt{2}, 3)^T, \quad y = (-3\sqrt{2}, 6)^T.$$

Multiplication gives $Ux = (3, -1, 1)^T$ and $Uy = (1, -7, 2)^T$. Consequently $(Ux)\cdot(Uy) = 3+7+2 = 12$, while $x \cdot y = (\sqrt{2})(-3\sqrt{2}) + 3(6) = -6 + 18 = 12$.

Conclusion. The two inner products agree exactly, although the outputs have three coordinates and the inputs only two.

Exercise 6.2.1: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 26 & 0 & 2 \\ 0 & 30 & 0 \\ 2 & 0 & 74 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.2: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 30 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

Exercise 6.2.3: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 126 & -30 \\ 0 & -30 & 11 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.4: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 38 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 56 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

The zero vector causes no difficulty for orthogonality, but it would prevent the set from being independent or orthonormal.

Exercise 6.2.5: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 23 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 122 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

Exercise 6.2.6: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 90 & -32 \\ 0 & -32 & 44 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.7: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^TU = \begin{bmatrix} 13 & 0 \\ 0 & 52 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{39}{13} = 3.$$

$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{26}{52} = \frac{1}{2}.$$

Conclusion. $x = (3)u_1 + (\frac{1}{2})u_2$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.8: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 40 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{-15}{10} = -\frac{3}{2}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{30}{40} = \frac{3}{4}.$$

Conclusion. $x = (-\frac{3}{2})u_1 + (\frac{3}{4})u_2$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.9: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^T U = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{5}{2} = \frac{5}{2}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{-27}{18} = -\frac{3}{2}.$$
$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{18}{9} = 2.$$

Conclusion. $x = (\frac{5}{2})u_1 + (-\frac{3}{2})u_2 + (2)u_3$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.10: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^TU = \begin{bmatrix} 18 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 18 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{24}{18} = \frac{4}{3}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{3}{9} = \frac{1}{3}.$$
$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{6}{18} = \frac{1}{3}.$$

Conclusion. $x = (\frac{4}{3})u_1 + (\frac{1}{3})u_2 + (\frac{1}{3})u_3$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.11: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 20 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{10}{20} = \frac{1}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -4 \\ 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.12: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 10 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-4}{10} = -\frac{2}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \begin{bmatrix} -\frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{2}{5} \\ -\frac{6}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{3}{5} \\ \frac{1}{5} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.13: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 65 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-13}{65} = -\frac{1}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 4 \\ -7 \end{bmatrix} \begin{bmatrix} -\frac{1}{5} \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} \\ \frac{7}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{14}{5} \\ \frac{8}{5} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.14: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 50 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{20}{50} = \frac{2}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} \begin{bmatrix} \frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{14}{5} \\ \frac{2}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{4}{5} \\ \frac{28}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.15: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 100 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{100} = \frac{3}{10}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} \begin{bmatrix} \frac{3}{10} \end{bmatrix} = \begin{bmatrix} \frac{12}{5} \\ \frac{9}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{1} = 1$, not the projection length.

Exercise 6.2.16: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 5 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{15}{5} = 3.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -6 \\ 3 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{45} = 3\sqrt{5}$, not the projection length.

Exercise 6.2.17: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.$$

The set is orthogonal but not normalized. Divide each column by its own positive length to get the following orthonormal columns:

$$Q = \begin{bmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & 0 \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} \end{bmatrix}, \qquad Q^T Q = I.$$

Exercise 6.2.18: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Conclusion. The set is not orthogonal and hence not orthonormal. Merely normalizing the individual lengths cannot make a nonzero pairwise dot product become zero.

Exercise 6.2.19: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.20: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 \\ 0 & \frac{5}{9} \end{bmatrix}.$$

The set is orthogonal but not normalized. Divide each column by its own positive length to get the following orthonormal columns:

$$Q = \begin{bmatrix} -\frac{2}{3} & \frac{\sqrt{5}}{5} \\ \frac{1}{3} & \frac{2\sqrt{5}}{5} \\ \frac{2}{3} & 0 \end{bmatrix}, \qquad Q^T Q = I.$$

Exercise 6.2.21: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.22: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.23: Orthogonality and orthogonal coordinates

(a) True. For example $(1, 0)^T, (1, 1)^T$ are independent but have dot product 1, so independence need not mean orthogonality.

(b) True. For a nonzero orthogonal family, take $y \cdot u_j$ to obtain $c_j = (y \cdot u_j)/(u_j \cdot u_j)$, without row reduction.

(c) False. Scaling two distinct vectors changes their dot product by a product of scalars; a zero product remains zero. Normalization uses nonzero scales and preserves orthogonality.

(d) False unless the matrix is square. The single column $(1, 0)^T$ is orthonormal but is a rectangular matrix, not an orthogonal matrix.

(e) False. The distance is $\|y - \widehat{y}\|$, not $\|\widehat{y}\|$. For nonzero $y \in L$, the distance is zero while $\|\widehat{y}\| = \|y\| > 0$.

Exercise 6.2.24: Orthogonality and orthogonal coordinates

(a) True. An orthogonal set can contain zero; $\{0, e_1\}$ is orthogonal but dependent. Nonzero orthogonal vectors are independent.

(b) False. The condition gives orthogonality only; the lengths may be different from one. For instance $\{2e_1, 3e_2\}$ meets it.

(c) True. $A^TA = I_n$ implies $\|Ax\|^2 = x^TA^TAx = x^Tx$. Taking nonnegative square roots gives length preservation.

(d) True. The factors of a nonzero c cancel in the projection formula, as proved explicitly in Exercise 31.

(e) True. Invertibility is part of the definition, and its inverse is its transpose.

Exercise 6.2.25: Orthogonality and orthogonal coordinates

Let U have orthonormal columns, so $U^TU = I$.

(b) Prove inner-product preservation first.

For all input vectors x, y, $(Ux) \cdot (Uy) = x^TU^TUy = x^Ty = x \cdot y$.

(a) Deduce preservation of lengths.

Set $y = x$ in the previous identity: $\|Ux\|^2 = \|x\|^2$. Since both norms are nonnegative, $\|Ux\| = \|x\|$.

(c) Deduce preservation of orthogonality in both directions.

Part (b) says that the two dot products are equal. Therefore

$$(Ux) \cdot (Uy) = 0 \quad \Longleftrightarrow \quad x \cdot y = 0.$$

This equivalence includes zero inputs, for which angle formulas would not be defined. As an additional consequence, for nonzero inputs the preserved dot products and norms also preserve the angle cosine.

Conclusion. All three conclusions of Theorem 7 follow from $U^TU = I$.

Exercise 6.2.26: Orthogonality and orthogonal coordinates

A nonzero orthogonal family is independent: in a zero combination, taking the dot product with each generator forces its coefficient to vanish. Thus the given n vectors are independent in $\mathbb{R}^n$. An independent set of exactly n vectors in this n-dimensional space is a basis, so its span is all of $\mathbb{R}^n$.

Conclusion. $W = \mathbb{R}^n$.

Exercise 6.2.27: Orthogonality and orthogonal coordinates

The orthonormal columns of U satisfy $U^TU = I_n$. Their nonzero orthogonality also makes them linearly independent. By the Invertible Matrix Theorem, a square matrix with independent columns is invertible. Multiplying $U^TU = I_n$ on the right by its inverse gives $U^T = U^{-1}$.

Conclusion. U is invertible and orthogonal. Squareness is essential to this inverse conclusion.

Exercise 6.2.28: Orthogonality and orthogonal coordinates

For an orthogonal square matrix, $U^{-1} = U^T$. Hence $UU^T = UU^{-1} = I_n$. The (i, j) entry of UU^T is the inner product of row i and row j. The identity matrix says these products are zero for distinct rows and one for each row with itself. Thus the rows are orthonormal. They are n nonzero orthogonal vectors in $\mathbb{R}^n$, so they form a basis.

Conclusion. The rows form an orthonormal basis, just as the columns do.

Exercise 6.2.29: Orthogonality and orthogonal coordinates

The product of invertible matrices is invertible and $(UV)^{-1} = V^{-1}U^{-1}$. Since U, V are orthogonal,

$$(UV)^{-1} = V^T U^T = (UV)^T,$$

using the transpose-of-product rule in the last equality.

Conclusion. UV is a square invertible matrix whose inverse is its transpose, so it is orthogonal.

Exercise 6.2.30: Orthogonality and orthogonal coordinates

A column permutation keeps the same column vectors, each of length one, with the same zero dot products between distinct columns. Thus V is still square with orthonormal columns. The result of Exercise 27 makes it invertible with inverse V^T. Equivalently write $V = UP$ for a permutation matrix P; then $V^T V = P^T U^T UP = P^T P = I$.

Conclusion. Every column permutation of an orthogonal matrix is orthogonal.

Exercise 6.2.31: Orthogonality and orthogonal coordinates

For $c \neq 0$, replace u by cu in the projection formula:

$$\frac{y \cdot (cu)}{(cu) \cdot (cu)}(cu) = \frac{c(y \cdot u)}{c^2(u \cdot u)}cu = \frac{y \cdot u}{u \cdot u}u.$$

The cancellation is valid because $c \neq 0$ and $u \neq 0$. Any two nonzero generators of a fixed line are related by just such a scale.

Conclusion. The projection depends on the line, not on the chosen nonzero generator or its orientation.

Exercise 6.2.32: Orthogonality and orthogonal coordinates

The two scaled vectors are nonzero because both original vectors and both scalars are nonzero. Their dot product is $(c_1 v_1) \cdot (c_2 v_2) = c_1 c_2 (v_1 \cdot v_2) = c_1 c_2 (0) = 0$. Thus they remain an orthogonal nonzero pair. Applying the same identity to each pair proves the assertion for a whole orthogonal family.

Conclusion. In particular normalization preserves orthogonality.

Exercise 6.2.33: Orthogonality and orthogonal coordinates

For any inputs x, y and scalars a, b,

$$\operatorname{proj}_L(ax+by) = \frac{(ax+by)\cdot u}{u\cdot u}u = a\frac{x\cdot u}{u\cdot u}u + b\frac{y\cdot u}{u\cdot u}u.$$

The right side is $a\operatorname{proj}_L x + b\operatorname{proj}_L y$, so both addition and scalar multiplication are preserved. The fixed nonzero u guarantees the denominator is positive.

Conclusion. Projection onto a fixed line through zero is linear.

Exercise 6.2.34: Orthogonality and orthogonal coordinates

Let $P(y) = \operatorname{proj}_L y$. Exercise 33 proves P linear; the identity map I is also linear. Reflection is $2P - I$. For scalars a, b,

$$(2P-I)(ay+bz) = 2aP(y) + 2bP(z) - ay - bz = a(2P(y)-y) + b(2P(z)-z).$$

Conclusion. Reflection in a fixed line through the origin is linear, with matrix $2uu^T/(u^Tu) - I$.

Exercise 6.2.35: Orthogonality and orthogonal coordinates

The appropriate calculation is the Gram matrix:

$$A^TA = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 \\ 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 100 \end{bmatrix}.$$

The six off-diagonal products are all zero, so the four columns are pairwise orthogonal. The diagonal entries also show that each column has length 10.

Conclusion. The columns are orthogonal, and their normalized matrix in the next exercise is $U = A/10$.

Exercise 6.2.36: Orthogonality and orthogonal coordinates

(a) Compute both products with the correct dimensions.

Normalization gives $U = A/10$, so

$$U^TU = I_4, \qquad UU^T = \frac{1}{100}\begin{bmatrix} 82 & 0 & -20 & 8 & 6 & 20 & 24 & 0 \\ 0 & 42 & 24 & 0 & -20 & 6 & 20 & -32 \\ -20 & 24 & 58 & 20 & 0 & 32 & 0 & 6 \\ 8 & 0 & 20 & 82 & 24 & -20 & 6 & 0 \\ 6 & -20 & 0 & 24 & 18 & 0 & -8 & 20 \\ 20 & 6 & 32 & -20 & 0 & 58 & 0 & 24 \\ 24 & 20 & 0 & 6 & -8 & 0 & 18 & -20 \\ 0 & -32 & 6 & 0 & 20 & 24 & -20 & 42 \end{bmatrix}.$$

The first product is 4×4 identity. The second is an 8×8 rank-four projection matrix, not I_8.

(b) Perform one reproducible random-vector calculation.

Integer draws from $[-3, 3]$ with seed 62036 give

$$y = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 0 \\ -1 \\ 2 \\ -3 \\ 0 \end{bmatrix}, \quad p = UU^T y = \begin{bmatrix} -1 \\ -\frac{47}{50} \\ \frac{1}{50} \\ -\frac{11}{10} \\ \frac{1}{5} \\ \frac{29}{50} \\ -\frac{9}{10} \\ \frac{57}{50} \end{bmatrix}, \quad z = y - p = \begin{bmatrix} 0 \\ -\frac{3}{50} \\ -\frac{51}{50} \\ \frac{11}{10} \\ -\frac{6}{5} \\ \frac{71}{50} \\ -\frac{21}{10} \\ -\frac{57}{50} \end{bmatrix}.$$

The formula $p = U(U^T y)$ explicitly writes p as a combination of the columns of U, so $p \in \operatorname{Col} U = \operatorname{Col} A$. Direct multiplication yields $z^T p = 0$.

(c) Check every column, not just p.

$$U^T z = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

These four entries are the four required column dot products.

(d) Extend the check to the whole column space.

If $w = Uc$, then $z \cdot w = z^T Uc = (U^T z)^T c = 0$. Thus $z \in (\operatorname{Col} A)^\perp$, and by construction $y = p + z$.

Conclusion. The displayed vectors give the requested orthogonal decomposition; the orthogonality also follows algebraically from $U^T(y - UU^T y) = 0$.

6.3 Orthogonal projections

For an orthogonal basis $u_1, \ldots, u_p$ of W, the orthogonal component of y in W is

$$\widehat{y} = \operatorname{proj}_W y = \sum_{j=1}^{p} \frac{y \cdot u_j}{u_j \cdot u_j} u_j.$$

The residual $z = y - \widehat{y}$ lies in $W^{\perp}$. Thus $y = \widehat{y} + z$ splits a vector into a part in the subspace and a perpendicular part. If the columns of U are *orthonormal*, this simplifies to $\widehat{y} = UU^T y$. The formula without the denominators is generally wrong for merely orthogonal columns of nonunit length.

For any $w \in W$, $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2$. Therefore $\widehat{y}$ is the unique closest point and the distance is $\|z\|$. Each numerical calculation below checks the residual against every generator, not just against the projection.

Practice problem 6.3.P1: Recognize a vector already in the subspace

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 66 & 0 \\ 0 & 6 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{88}{66} = \frac{4}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-2}{6} = -\frac{1}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -7 & -1 \\ 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The projection equals $y = (-9, 1, 6)^T$ itself; the residual is zero.

Exercise 6.3.1: Compute and check an orthogonal projection

Use the orthogonal basis formula for all four coefficients:

$$c_j = \frac{x \cdot u_j}{u_j \cdot u_j}, \qquad (c_1, c_2, c_3, c_4) = (-\frac{8}{9}, -\frac{2}{9}, \frac{2}{3}, 2).$$

The fourth component and the sum of the first three components are

$$q = c_4u_4 = \begin{bmatrix} 10 \\ -6 \\ -2 \\ 2 \end{bmatrix}, \qquad p = x - q = \begin{bmatrix} 0 \\ -2 \\ 4 \\ -2 \end{bmatrix}.$$

Here $p = c_1u_1 + c_2u_2 + c_3u_3$ is in the first requested span and q is in the second. The basis orthogonality makes $p \cdot q = 0$.

Conclusion. $x = p + q$ with the two explicit columns above.

Exercise 6.3.2: Compute and check an orthogonal projection

For the four orthogonal generators, compute $c_j = (v \cdot u_j)/(u_j \cdot u_j)$. The exact coefficients are

$$(c_1, c_2, c_3, c_4) = (2, \frac{3}{7}, \frac{12}{7}, -\frac{8}{7}).$$

Thus the requested split is

$$p = c_1u_1 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 2 \end{bmatrix}, \qquad q = c_2u_2 + c_3u_3 + c_4u_4 = \begin{bmatrix} 2 \\ 1 \\ -5 \\ 1 \end{bmatrix}.$$

Adding the two columns gives the original v. The first is in $\text{Span}\{u_1\}$ and the second in $\text{Span}\{u_2, u_3, u_4\}$, and they are orthogonal.

Conclusion. $v = p + q$ as displayed.

Exercise 6.3.3: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{3}{2} = \frac{3}{2}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{5}{2} = \frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{2} \\ \frac{5}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.4: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{25} = \frac{6}{5}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-15}{25} = -\frac{3}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & -4 \\ 4 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{6}{5} \\ -\frac{3}{5} \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.5: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 14 & 0 \\ 0 & 6 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{7}{14} = \frac{1}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-15}{6} = -\frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & 1 \\ -1 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{5}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.6: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 18 & 0 \\ 0 & 2 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-27}{18} = -\frac{3}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{5}{2} = \frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -4 & 0 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{3}{2} \\ \frac{5}{2} \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.7: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 14 & 0 \\ 0 & 42 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{0}{14} = 0.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{28}{42} = \frac{2}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 3 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{2}{3} \\ \frac{8}{3} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{7}{3} \\ \frac{7}{3} \\ \frac{7}{3} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.8: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 \\ 0 & 14 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{3} = 2.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{7}{14} = \frac{1}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 \\ 1 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{7}{2} \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{5}{2} \\ \frac{1}{2} \\ 2 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.9: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{3} = 2.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{10}{15} = \frac{2}{3}.$$
$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = \frac{-2}{3} = -\frac{2}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ -1 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.10: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{1}{3} = \frac{1}{3}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{14}{3} = \frac{14}{3}.$$
$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = \frac{-5}{3} = -\frac{5}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ -\frac{5}{3} \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -2 \\ 2 \\ 2 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.11: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 12 & 0 \\ 0 & 4 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{12} = \frac{1}{2}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{6}{4} = \frac{3}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 2 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.12: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 26 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{10} = 3.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{26}{26} = 1.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -4 \\ -2 & 1 \\ -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.13: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 15 & 0 \\ 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{10}{15} = \frac{2}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-7}{3} = -\frac{7}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \\ -3 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ -\frac{7}{3} \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ -4 \\ 4 \\ 0 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.14: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 14 & 0 \\ 0 & 49 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{7}{14} = \frac{1}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{0}{49} = 0.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 & 5 \\ 0 & -2 \\ -1 & 4 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ -\frac{3}{2} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 1 \\ 4 \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.15: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 35 & 0 \\ 0 & 14 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{35}{35} = 1.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-28}{14} = -2.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -3 & -3 \\ -5 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \\ -1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{40} = 2\sqrt{10}$, not the projection length.

Exercise 6.3.16: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 26 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{10} = 3.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{26}{26} = 1.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -4 \\ -2 & 1 \\ -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{64} = 8$, not the projection length.

Exercise 6.3.17: Compute and check an orthogonal projection

(a) Compute the two distinct products.

$$U^T U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad UU^T = \begin{bmatrix} \frac{8}{9} & -\frac{2}{9} & \frac{2}{9} \\ -\frac{2}{9} & \frac{5}{9} & \frac{4}{9} \\ \frac{2}{9} & \frac{4}{9} & \frac{5}{9} \end{bmatrix}.$$

The first product checks orthonormal columns. The second acts on the ambient space and projects onto their span.

(b) Compute by the orthonormal projection formula.

$$\operatorname{proj}_W y = U(U^T y) = (UU^T)y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}.$$

Both methods are the same matrix product by associativity. Directly $U^T(y - \operatorname{proj}_W y) = 0$, checking the residual.

Exercise 6.3.18: Compute and check an orthogonal projection

(a) Compute the two distinct products.

$$U^T U = \begin{bmatrix} 1 \end{bmatrix}, \qquad UU^T = \begin{bmatrix} \frac{1}{10} & -\frac{3}{10} \\ -\frac{3}{10} & \frac{9}{10} \end{bmatrix}.$$

The first product checks orthonormal columns. The second acts on the ambient space and projects onto their span.

(b) Compute by the orthonormal projection formula.

$$\operatorname{proj}_W y = U(U^T y) = (UU^T)y = \begin{bmatrix} -2 \\ 6 \end{bmatrix}.$$

Both methods are the same matrix product by associativity. Directly $U^T(y - \operatorname{proj}_W y) = 0$, checking the residual.

Exercise 6.3.19: Compute and check an orthogonal projection

Plan. Remove the component of the given outside vector that lies in W; what remains is perpendicular to W.

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-2}{6} = -\frac{1}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{2}{30} = \frac{1}{15}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{15} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{2}{5} \\ \frac{4}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ \frac{2}{5} \\ \frac{1}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. Take $v = \begin{bmatrix} 0 \\ \frac{2}{5} \\ \frac{1}{5} \end{bmatrix}$. It is nonzero (its displayed entries are not all zero) and has zero dot product with u_1 and u_2.

Exercise 6.3.20: Compute and check an orthogonal projection

Plan. Remove the component of the given outside vector that lies in W; what remains is perpendicular to W.

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{1}{6} = \frac{1}{6}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-1}{30} = -\frac{1}{30}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{30} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{5} \\ -\frac{2}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. Take $v = \begin{bmatrix} 0 \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix}$. It is nonzero (its displayed entries are not all zero) and has zero dot product with u_1 and u_2.

Exercise 6.3.21: Compute and check an orthogonal projection

(a) True. For any $w = c_1 u_1 + c_2 u_2$, $z \cdot w = c_1(z \cdot u_1) + c_2(z \cdot u_2) = 0$.

(b) True. This is the residual conclusion of the Orthogonal Decomposition Theorem.

(c) False. Orthogonal decomposition into W and $W^{\perp}$ is unique, so any orthogonal basis of the same W gives the same projection.

(d) True. For $y \in W$, the decomposition $y = y + 0$ already has its first part in W and its second in $W^{\perp}$, so uniqueness applies.

(e) True. With orthonormal columns, each projection coefficient is $u_j \cdot y$; collecting the coefficients gives $U(U^T y)$.

Exercise 6.3.22: Compute and check an orthogonal projection

(a) True. Such a vector is orthogonal to itself, so its squared length is zero and it must be zero.

(b) True. Each term $((y \cdot u_j)/(u_j \cdot u_j))u_j$ is the orthogonal projection onto the one-dimensional subspace $\text{Span}\{u_j\}$.

(c) True. The assumed sum is an orthogonal decomposition into W and $W^{\perp}$, and its uniqueness identifies the first part.

(d) False. The closest point is $\text{proj}_W y$; the stated vector is the residual. For a nonzero $y \in W$, the stated vector is zero while the unique best approximation is y.

(e) False. UU^T fixes only the vectors in $\text{Col}\, U$ in general. For $U = (1, 0)^T$ and $x = (0, 1)^T$, $UU^T x = 0 \neq x$.

Exercise 6.3.23: Compute and check an orthogonal projection

Apply orthogonal decomposition to the row space.

Theorem 3 identifies $(\text{Row}\, A)^{\perp} = \text{Nul}\, A$. Taking $W = \text{Row}\, A$ in the decomposition theorem gives $x = p + u$ with $p \in \text{Row}\, A$ and $u \in \text{Nul}\, A$.

Existence of a row-space solution for a consistent equation.

If $Ax = b$ has a solution x, decompose that solution as above. Since $Au = 0$, we obtain $Ap = A(x - u) = b$. Thus some row-space vector does solve the equation.

Uniqueness within the row space.

If p, p' are row-space solutions, then $A(p - p') = 0$. Their difference lies both in the row space and its perpendicular null space, so it is zero. Therefore $p = p'$.

Conclusion. Every vector has the required split, and every consistent system has exactly one solution belonging to the row space.

Exercise 6.3.24: Compute and check an orthogonal projection

(a) Check every type of pair.

Two distinct w_i are orthogonal by assumption, as are two distinct v_j. A mixed pair is orthogonal because $w_i \in W$ and $v_j \in W^\perp$. Thus the concatenated family is orthogonal. Its members are all nonzero because they belong to bases.

(b) Span an arbitrary ambient vector.

Every $x \in \mathbb{R}^n$ splits as $x = w + v$, where $w \in W$ and $v \in W^\perp$. Expand w in the w_i basis and v in the v_j basis. Adding the expansions expresses x using the concatenated family.

(c) Count the resulting basis.

A nonzero orthogonal family is independent, and (b) proves spanning. Hence this family is a basis of $\mathbb{R}^n$ with $p + q$ members.

Conclusion. $\dim W + \dim W^\perp = p + q = n$. This includes the zero-space boundary cases, where one of the bases is empty.

Exercise 6.3.25: Compute and check an orthogonal projection

Use the normalized matrix from Exercise 6.2.36.

That matrix is $U = A/10$, with the explicitly supplied 8×4 integer matrix A from Exercise 6.2.35. It satisfies $U^T U = I_4$.

$$\widehat{y} = U(U^T y) = \begin{bmatrix} \frac{6}{5} \\ \frac{3}{5} \\ \frac{1}{5} \\ \frac{2}{5} \\ \frac{6}{5} \\ \frac{3}{5} \\ \frac{1}{5} \\ \frac{2}{5} \end{bmatrix}, \qquad y - \widehat{y} = \begin{bmatrix} -\frac{1}{5} \\ \frac{3}{5} \\ -\frac{1}{5} \\ -\frac{1}{5} \\ \frac{3}{5} \\ -\frac{1}{5} \\ \frac{3}{5} \\ \frac{3}{5} \end{bmatrix}.$$

Commands reproducing the exact calculation.

```
from sympy import Matrix, ones
A = Matrix([[-6,-3,6,1],[-1,2,1,-6],[3,6,3,-2],
            [6,-3,6,-1],[2,-1,2,3],[-3,6,3,2],
            [-2,-1,2,-3],[1,2,1,6]])
U = A/10
y = ones(8,1)
p = U*(U.T*y)
print(p)
assert U.T*(y-p) == Matrix.zeros(4,1)
```

The zero dot-product column $U^T(y - p) = 0$ checks that the residual is perpendicular to the subspace, proving that the displayed point is its projection and hence the unique closest point.

Exercise 6.3.26: Compute and check an orthogonal projection

Use the normalized matrix from Exercise 6.2.36.

That matrix is $U = A/10$, with the explicitly supplied 8×4 integer matrix A from Exercise 6.2.35. It satisfies $U^T U = I_4$.

$$\widehat{y} = U(U^T y) = \begin{bmatrix} \frac{1}{5} \\ \frac{23}{25} \\ \frac{11}{25} \\ 1 \\ -\frac{1}{5} \\ -\frac{11}{25} \\ \frac{3}{5} \\ -\frac{23}{25} \end{bmatrix}, \qquad y - \widehat{y} = \begin{bmatrix} \frac{4}{5} \\ \frac{2}{25} \\ \frac{14}{25} \\ 0 \\ -\frac{4}{5} \\ -\frac{14}{25} \\ -\frac{8}{5} \\ -\frac{2}{25} \end{bmatrix}.$$

Conclusion. The distance is $\|y - \widehat{y}\| = \sqrt{\frac{112}{25}} = \frac{4\sqrt{7}}{5}$. Here y denotes the exercise target b.

6.4 The Gram–Schmidt process

Gram–Schmidt replaces an independent ordered list by orthogonal directions with the same successive spans. At stage k, subtract the projections of x_k onto every earlier orthogonal direction. Independence guarantees the residual is nonzero; normalizing it gives the next unit direction.

For a full-column-rank matrix A, the normalized columns form Q with $Q^TQ = I$. The factor $R = Q^TA$ is square upper triangular and satisfies $A = QR$. Its diagonal can be made positive by using positive lengths in the normalizations. In the calculations below we verify both $Q^TQ = I$ and $QR = A$, not merely the shape of the proposed factors.

Practice problem 6.4.P1: Normalize a pair that is already orthogonal

The two given vectors satisfy $(1,1,1)\cdot(1/3,1/3,-2/3) = 0$. Their lengths are $\sqrt{3}$ and $\sqrt{6}/3$, respectively. Hence an orthonormal basis is

$$q_1 = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad q_2 = \frac{1}{\sqrt{6}}(1,1,-2)^T.$$

Each has squared length one, their dot product is zero, and their nonzero rescalings span the same space as the original pair.

Conclusion. The displayed q_1, q_2 are the desired orthonormal basis.

Exercise 6.4.1: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & 8 \\ 0 & 5 \\ -1 & -6 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (3).$$

Subtract all of those components:

$$v_2 = x_2 - (3)w_1 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 10 & 0 \\ 0 & 35 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.2: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 0 & 5 \\ 4 & 6 \\ 2 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (1).$$

Subtract all of those components:

$$v_2 = x_2 - (1)w_1 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 \\ 0 & 105 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.3: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 2 & 4 \\ -5 & -1 \\ 1 & 2 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}).$$

Subtract all of those components:

$$v_2 = x_2 - (\frac{1}{2})w_1 = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Choose $w_2 = (\frac{2}{3})v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 30 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.4: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -3 \\ -4 & 14 \\ 5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 50 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.5: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 7 \\ -4 & -7 \\ 0 & -4 \\ 1 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (2).$$

Subtract all of those components:

$$v_2 = x_2 - (2)w_1 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 18 & 0 \\ 0 & 43 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.6: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -5 \\ -1 & 9 \\ 2 & -9 \\ -1 & 3 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-3).$$

Subtract all of those components:

$$v_2 = x_2 - (-3)w_1 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 15 & 0 \\ 0 & 61 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.7: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 2 & 4 \\ -5 & -1 \\ 1 & 2 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}).$$

Subtract all of those components:

$$v_2 = x_2 - (\frac{1}{2})w_1 = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Choose $w_2 = (\frac{2}{3})v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 30 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{30}}{15} & \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{30}}{6} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{30}}{30} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Conclusion. The columns of Q are an orthonormal basis for the original column span.

Exercise 6.4.8: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -3 \\ -4 & 14 \\ 5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 50 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \dfrac{w_1}{\|w_1\|} & \cdots & \dfrac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{2}}{10} & \frac{\sqrt{6}}{6} \\ -\frac{2\sqrt{2}}{5} & \frac{\sqrt{6}}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Conclusion. The columns of Q are an orthonormal basis for the original column span.

Exercise 6.4.9: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{3}{2}, -\frac{1}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{3}{2})w_1 - (-\frac{1}{2})w_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}.$$

Choose $w_3 = v_3 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.10: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -1 & 6 & 6 \\ 3 & -8 & 3 \\ 1 & -2 & 6 \\ 1 & -4 & -3 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-3).$$

Subtract all of those components:

$$v_2 = x_2 - (-3)w_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}, \frac{5}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{1}{2})w_1 - (\frac{5}{2})w_2 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}.$$

Choose $w_3 = v_3 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.11: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (-1).$$

Subtract all of those components:

$$v_2 = x_2 - (-1)w_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (4, -1).$$

Subtract all of those components:

$$v_3 = x_3 - (4)w_1 - (-1)w_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{2})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.12: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (4).$$

Subtract all of those components:

$$v_2 = x_2 - (4)w_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (\frac{7}{2}, \frac{3}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{7}{2})w_1 - (\frac{3}{2})w_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{3})v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.13: Gram–Schmidt with every component removed

Use the known orthonormal factor to isolate R.

The supplied Q satisfies $Q^T Q = I_2$. Left multiplication of $A = QR$ by Q^T gives $R = Q^T A$, so

$$R = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}.$$

Verify the matrix product, not just triangularity.

$$Q = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ \frac{1}{6} & \frac{5}{6} \\ -\frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} \end{bmatrix}, \qquad QR = \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = A.$$

Conclusion. The displayed upper triangular R is the required factor.

Exercise 6.4.14: Gram–Schmidt with every component removed

Use the known orthonormal factor to isolate R.

The supplied Q satisfies $Q^TQ = I_2$. Left multiplication of $A = QR$ by Q^T gives $R = Q^TA$, so

$$R = \begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix}.$$

Verify the matrix product, not just triangularity.

$$Q = \begin{bmatrix} -\frac{2}{7} & \frac{5}{7} \\ \frac{5}{7} & \frac{2}{7} \\ \frac{2}{7} & -\frac{4}{7} \\ \frac{4}{7} & \frac{2}{7} \end{bmatrix}, \qquad QR = \begin{bmatrix} -2 & 3 \\ 5 & 7 \\ 2 & -2 \\ 4 & 6 \end{bmatrix} = A.$$

Conclusion. The displayed upper triangular R is the required factor.

Exercise 6.4.15: An exact QR factorization and its checks

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (-1).$$

Subtract all of those components:

$$v_2 = x_2 - (-1)w_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (4, -1).$$

Subtract all of those components:

$$v_3 = x_3 - (4)w_1 - (-1)w_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{2})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{5}}{5} & \frac{1}{2} & \frac{1}{2} \\ -\frac{\sqrt{5}}{5} & 0 & 0 \\ -\frac{\sqrt{5}}{5} & \frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{5}}{5} & -\frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{5}}{5} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

The exact check $Q^TQ = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^TA = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.16: An exact QR factorization and its checks

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (4).$$

Subtract all of those components:

$$v_2 = x_2 - (4)w_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{7}{2}, \frac{3}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{7}{2})w_1 - (\frac{3}{2})w_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{3})v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{2}}{4} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{2}}{4} & \frac{1}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{2}}{4} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{2}}{4} & \frac{1}{2} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^T A = \begin{bmatrix} 2 & 8 & 7 \\ 0 & 2\sqrt{2} & 3\sqrt{2} \\ 0 & 0 & 6 \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.17: Gram–Schmidt with every component removed

(a) False as stated: $c = 0$ replaces a nonzero basis vector by zero, which makes the family dependent and unable to retain its old span. For $c \neq 0$ it would remain an orthogonal basis.

(b) True. Each residual is obtained by subtracting a combination of earlier directions, and the reverse equation reconstructs the new input. Induction proves equality of each pair of successive spans.

(c) True. Since $Q^T Q = I$, left multiplication of $A = QR$ gives $Q^T A = Q^T QR = R$.

Exercise 6.4.18: Gram–Schmidt with every component removed

(a) False without the requirement that all v_i are nonzero. In a three-dimensional W, the set $\{e_1, e_2, 0\}$ is orthogonal but is not a basis. If the three orthogonal vectors were all nonzero, they would be independent and would form a basis.

(b) True. If $x - \text{proj}_W x = 0$, then $x = \text{proj}_W x \in W$, contradicting $x \notin W$.

(c) True. Gram–Schmidt produces orthonormal columns of Q with exactly the same column span as A; they therefore form a basis of that column space.

Exercise 6.4.19: Gram–Schmidt with every component removed

Suppose $Rx = 0$. Multiply by Q to get $Ax = QRx = 0$. Independence of the columns of A means its homogeneous system has only the zero solution, so $x = 0$. Thus the square $n \times n$ matrix R has trivial null space. By the Invertible Matrix Theorem, it is invertible.

Conclusion. The conclusion requires neither guessing a determinant nor assuming R is already invertible; it follows from independence of A.

Exercise 6.4.20: Gram–Schmidt with every component removed

If $y \in \text{Col}\, A$, write $y = Ax = Q(Rx)$. This represents y as a combination of columns of Q, proving $\text{Col}\, A \subseteq \text{Col}\, Q$. Conversely, if $y \in \text{Col}\, Q$, write $y = Qz$ and set $x = R^{-1}z$. Then $Ax = QR(R^{-1}z) = Qz = y$, so $y \in \text{Col}\, A$.

Conclusion. The two inclusions give $\text{Col}\, A = \text{Col}\, Q$. Invertibility of R is needed for the reverse inclusion.

Exercise 6.4.21: Gram–Schmidt with every component removed

Start with the thin factorization $A = QR$, where Q has n orthonormal columns in $\mathbb{R}^m$ and R is invertible upper triangular. Extend those columns to an orthonormal basis of $\mathbb{R}^m$ by taking an orthonormal basis $Q_\perp$ of $(\operatorname{Col} Q)^\perp$; for example append independent standard coordinate directions and continue Gram–Schmidt. Set $Q_1 = [Q\ Q_\perp]$. Its m columns are orthonormal, so it is an orthogonal $m \times m$ matrix. Then

$$Q_1 \begin{bmatrix} R \\ 0_{(m-n)\times n} \end{bmatrix} = QR + Q_\perp 0 = A.$$

Conclusion. Adjoining the orthonormal complement produces the full QR factorization. The triangular factor in the top block remains the same R.

Exercise 6.4.22: Gram–Schmidt with every component removed

Using the orthogonal projection formula and nonzero basis vectors,

$$T(x) = \sum_{j=1}^{p} \frac{x \cdot u_j}{u_j \cdot u_j} u_j.$$

For any x, y and scalars a, b, substitute $ax+by$ and distribute each numerator. The fixed denominators give

$$T(ax + by) = \sum_j \frac{a(x \cdot u_j) + b(y \cdot u_j)}{u_j \cdot u_j} u_j = aT(x) + bT(y).$$

Conclusion. Projection onto a fixed subspace is linear.

Exercise 6.4.23: Gram–Schmidt with every component removed

Partition $Q = [Q_1\ Q_2]$ after its first p columns and partition the upper triangular matrix R compatibly:

$$R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}.$$

The first p columns of QR are therefore $Q_1 R_{11}$, so $A_1 = Q_1 R_{11}$. A subset of the orthonormal columns of Q remains orthonormal, and R_{11} is upper triangular. Its diagonal entries are the first p nonzero diagonal entries of R, so it is invertible.

Conclusion. Retain the first p columns of Q and the leading $p \times p$ block of R to obtain a QR factorization of A_1.

Exercise 6.4.24: Four checked orthogonal directions in five coordinates

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -10 \\ 2 \\ -6 \\ 16 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} -5 \\ 1 \\ -3 \\ 8 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (-1, -4).$$

Subtract all of those components:

$$v_3 = x_3 - (-1)w_1 - (-4)w_2 = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{6})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Column 4.

The component coefficients along the previous w_j are

$$\frac{x_4 \cdot w_j}{w_j \cdot w_j} : \quad (1, -3, -3).$$

Subtract all of those components:

$$v_4 = x_4 - (1)w_1 - (-3)w_2 - (-3)w_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}.$$

Choose $w_4 = (\frac{1}{5})v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.25: QR factors for the five-by-four matrix

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -10 \\ 2 \\ -6 \\ 16 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} -5 \\ 1 \\ -3 \\ 8 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (-1, -4).$$

Subtract all of those components:

$$v_3 = x_3 - (-1)w_1 - (-4)w_2 = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{6})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Column 4.

The component coefficients along the previous w_j are

$$\frac{x_4 \cdot w_j}{w_j \cdot w_j}: \quad (1, -3, -3).$$

Subtract all of those components:

$$v_4 = x_4 - (1)w_1 - (-3)w_2 - (-3)w_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}.$$

Choose $w_4 = (\frac{1}{5})v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & \frac{\sqrt{2}}{2} \\ -\frac{3}{10} & -\frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{4}{5} & 0 & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^T A = \begin{bmatrix} 20 & -20 & -10 & 10 \\ 0 & 6 & -8 & -6 \\ 0 & 0 & 6\sqrt{3} & -3\sqrt{3} \\ 0 & 0 & 0 & 5\sqrt{2} \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.26: Gram–Schmidt with every component removed

Implement the specified orthonormal update in exact arithmetic.

```
from sympy import Matrix, zeros, sqrt, simplify
A = Matrix([[-10,13,7,-11],[2,1,-5,3],[-6,3,13,-3],
            [16,-16,-2,5],[2,1,-5,-7]])
Q = zeros(A.rows,0)
for k in range(A.cols):
    x = A[:,k]
    v = simplify(x-Q*(Q.T*x))
    q = simplify(v/sqrt(v.dot(v)))
    Q = Q.row_join(q)
R = simplify(Q.T*A)
assert simplify(Q.T*Q) == Matrix.eye(4)
assert simplify(Q*R) == A
print(Q)
print(R)
```

Record the actual computed factors.

$$Q = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & \frac{\sqrt{2}}{2} \\ -\frac{3}{10} & -\frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{4}{5} & 0 & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}, \qquad R = \begin{bmatrix} 20 & -20 & -10 & 10 \\ 0 & 6 & -8 & -6 \\ 0 & 0 & 6\sqrt{3} & -3\sqrt{3} \\ 0 & 0 & 0 & 5\sqrt{2} \end{bmatrix}.$$

The update subtracts all previous components at once through $Q(Q^T x)$. Nonzero residuals follow from independent input columns. These exact factors agree with Exercise 25; the two displayed assertions verify unit orthogonality and reconstruction.

Conclusion. The commands implement the requested recurrence and produce the explicit checked QR factorization above.

6.5 Least-squares problems

A least-squares solution $\widehat{x}$ minimizes the *output error* $\|b - Ax\|$, not $\|b - x\|$. The minimizing output $\widehat{b} = A\widehat{x}$ is the orthogonal projection of b onto $\operatorname{Col} A$. The residual $r = b - A\widehat{x}$ is orthogonal to every column, so

$$A^T r = 0 \quad \Longleftrightarrow \quad A^T A\widehat{x} = A^T b.$$

These are the normal equations. They always have solutions, but the coefficient vector need not be unique when A has dependent columns. The projection is still unique. With full column rank and a QR factorization $A = QR$, solve $R\widehat{x} = Q^T b$ by back substitution; do not explicitly form an inverse for numerical work.

Practice problem 6.5.P1: A least-squares problem can have zero error

For the given matrix, direct multiplication gives

$$A(2, -1, 0)^T = (5, -3, -5)^T = b.$$

Thus $\widehat{x} = (2, -1, 0)^T$ is already an exact solution and therefore a least-squares solution: its residual is zero, and no norm can be smaller than zero. Its associated least-squares error is 0. For a normal-equation check,

$$A^T A = \begin{bmatrix} 3 & 9 & 0 \\ 9 & 83 & 28 \\ 0 & 28 & 14 \end{bmatrix}, \quad A^T b = (-3, -65, -28)^T,$$

and the chosen vector satisfies that system as well.

Practice problem 6.5.P2: An orthogonal target has zero as its projection

If b is orthogonal to all columns, $A^T b = 0$. The projection of b onto $\operatorname{Col} A$ is zero, so $x = 0$ is a least-squares solution and the minimum error is $\|b\|$. More precisely all least-squares solutions satisfy $Ax = 0$; thus they form $\operatorname{Nul} A$. The coefficient vector is unique only when that null space is $\{0\}$.

Exercise 6.5.1: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} -4 \\ 11 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^TA \mid A^Tb] \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}.$$

Substituting into A^Tr gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.2: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}.$$

Taking column dot products gives

$$A^TA = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} -24 \\ -2 \end{bmatrix}.$$

The normal equation is $A^TA\widehat{x} = A^Tb$ with these entries.

(b) Solve that smaller system.

$$[A^TA \mid A^Tb] \sim \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 3 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Substituting into A^Tr gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.3: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 6 \\ -6 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^T A \mid A^T b] \sim \begin{bmatrix} 1 & 0 & \frac{4}{3} \\ 0 & 1 & -\frac{1}{3} \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}.$$

Substituting into $A^T r$ gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.4: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 6 \\ 14 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^T A \mid A^T b] \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}.$$

Substituting into $A^T r$ gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.5: Find every minimizing coefficient vector

The original data give

$$A^T A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}.$$

The reduced augmented normal matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Set the free variable $x_3 = t$. Reading the first two equations gives

$$\widehat{x} = \begin{bmatrix} 5 - t \\ t - 3 \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

All these coefficient vectors produce the same fitted output $A\widehat{x} = \begin{bmatrix} 2 \\ 2 \\ 5 \\ 5 \end{bmatrix}$. The residual $\begin{bmatrix} -1 \\ 1 \\ 3 \\ -3 \end{bmatrix}$ is orthogonal to all columns of A. Conversely every minimizer solves the normal equations, so this family contains all least-squares solutions, not just selected ones.

Exercise 6.5.6: Find every minimizing coefficient vector

The original data give

$$A^T A = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 3 & 0 \\ 3 & 0 & 3 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 27 \\ 12 \\ 15 \end{bmatrix}.$$

The reduced augmented normal matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Set the free variable $x_3 = t$. Reading the first two equations gives

$$\widehat{x} = \begin{bmatrix} 5 - t \\ t - 1 \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

All these coefficient vectors produce the same fitted output $A\widehat{x} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 5 \\ 5 \\ 5 \end{bmatrix}$. The residual $\begin{bmatrix} 3 \\ -2 \\ -1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$ is orthogonal to all columns of A. Conversely every minimizer solves the normal equations, so this family contains all least-squares solutions, not just selected ones.

Exercise 6.5.7: Compute the minimum error as a residual norm

Using the least-squares solution in Exercise 3, the residual is $r = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$. Therefore

$$\|b - A\widehat{x}\| = \sqrt{(1)^2 + (3)^2 + (-3)^2 + (1)^2} = \boxed{2\sqrt{5}}.$$

The error is the norm, rather than its square. The sum of squared residuals is 20.

Exercise 6.5.8: Compute the minimum error as a residual norm

Using the least-squares solution in Exercise 4, the residual is $r = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. Therefore

$$\|b - A\widehat{x}\| = \sqrt{(1)^2 + (1)^2 + (-2)^2} = \boxed{\sqrt{6}}.$$

The error is the norm, rather than its square. The sum of squared residuals is 6.

Exercise 6.5.9: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^T A = \begin{bmatrix} 14 & 0 \\ 0 & 42 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{2}{7} \\ \frac{1}{7} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.10: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 24 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 9 \\ 12 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.11: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^TA = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} 36 \\ 0 \\ 9 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^Tb)/(a_j^Ta_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ -1 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{1}{3} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} 6 \\ -1 \\ -4 \\ 1 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.12: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^TA = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} 1 \\ 14 \\ -5 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^Tb)/(a_j^Ta_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ -\frac{5}{3} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} -3 \\ 3 \\ 3 \\ 0 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.13: Disqualify one candidate by finding a better one

Direct multiplication gives $Au = (11, -11, 11)^T$ and $Av = (7, -12, 7)^T$. With $b = (11, -9, 5)^T$, the residuals are $(0, 2, -6)^T$ and $(4, 3, -2)^T$, respectively. Their squared norms are 40 and 29. Since the candidate v has strictly smaller error than u, the vector u cannot minimize the error. This proves u is not a least-squares solution without solving the least-squares problem. It does not assert that v is a minimizer.

Exercise 6.5.14: Equal candidate errors do not prove minimality

Multiplication gives $Au = (3, 8, 2)^T$, $Av = (7, 2, 8)^T$. The two residuals from $b = (5, 4, 4)^T$ are $(2, -4, 2)^T$ and $(-2, 2, -4)^T$, both with squared norm 24. Now take the midpoint $z = (u + v)/2 = (5, -5)^T$. Linearity gives $Az = (5, 5, 5)^T$, whose residual is $(0, -1, -1)^T$, with squared norm $2 < 24$. Thus *neither* u nor v can be a least-squares solution. Exhibiting this one better candidate suffices; we have not computed the actual minimizer.

Exercise 6.5.15: Use the supplied QR factorization

The columns of Q are orthonormal, so solve $R\widehat{x} = Q^T b$. Here

$$Q^T b = \begin{bmatrix} (14 + 6 + 1)/3 \\ (-7 + 6 - 2)/3 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 5 \\ 0 & 1 \end{bmatrix}.$$

Back substitution first gives $\widehat{x}_2 = -1$, then $3\widehat{x}_1 + 5(-1) = 7$, so $\widehat{x}_1 = 4$.

Conclusion. $\widehat{x} = (4, -1)^T$. No explicit inverse of R is needed.

Exercise 6.5.16: Back substitute in another triangular system

The given columns of Q give

$$Q^T b = \begin{bmatrix} (-1 + 6 + 5 + 7)/2 \\ (1 + 6 - 5 + 7)/2 \end{bmatrix} = \begin{bmatrix} 17/2 \\ 9/2 \end{bmatrix}.$$

Thus $5\widehat{x}_2 = 9/2$ and $2\widehat{x}_1 + 3\widehat{x}_2 = 17/2$. These yield $\widehat{x}_2 = 9/10$, $\widehat{x}_1 = 29/10$.

Conclusion. $\widehat{x} = (2.9, 0.9)^T$, the unique least-squares solution.

Exercise 6.5.17: Five least-squares assertions

(a) **True.** This is the defining minimization problem in the Euclidean output norm.

(b) **True.** The nearest output in $\operatorname{Col} A$ is precisely $\widehat{b} = \operatorname{proj}_{\operatorname{Col} A} b$.

(c) **False.** The displayed inequality is reversed. A minimizer satisfies $\|b - A\widehat{x}\| \leq \|b - Ax\|$ for every x, not the other way around.

(d) **True.** The normal equations are equivalent to $A^T(b - A\widehat{x}) = 0$, the orthogonal-projection condition.

(e) **True.** Independent columns make A^TA invertible, so the always-consistent normal equations have exactly one solution.

Exercise 6.5.18: Distinguish input coefficients from fitted outputs

(a) **True.** An exact solution has zero error, the smallest possible.

(b) **False.** The point in $\operatorname{Col} A$ closest to b is $A\widehat{x}$, not generally $\widehat{x}$, which has n input coordinates rather than m output coordinates.

(c) **True.** The entries of $\widehat{x}$ are the weights whose column combination produces the projected output.

(d) **False without a rank assumption.** A^TA may be singular, so the displayed inverse may not exist. For the zero matrix every input is a minimizer, but A^TA is not invertible in positive dimension.

(e) **False as a numerical claim.** In exact arithmetic they characterize minimizers, but forming A^TA can magnify conditioning problems and roundoff. QR methods are often more reliable.

(f) **False as stated about the best computation.** With full column rank, the formula is algebraically valid, but solving $R\widehat{x} = Q^Tb$ by back substitution is preferable to explicitly forming R^{-1}. If full column rank is absent, the stated inverse may not exist at all.

Exercise 6.5.19: Prove equality of two null spaces

(a) Start with $Ax = 0$.

Multiplying by A^T gives $A^TAx = A^T0 = 0$.

(b) Start with $A^TAx = 0$.

Multiplying on the left by x^T gives $0 = x^TA^TAx = (Ax)^T(Ax) = \|Ax\|^2$. A sum of squares is zero only when every entry is zero, so $Ax = 0$.

Conclusion. Both inclusions hold, hence $\operatorname{Nul} A = \operatorname{Nul}(A^TA)$.

Exercise 6.5.20: Invertibility of the Gram matrix forces independent columns

If $Ax = 0$, Exercise 19 gives $A^TAx = 0$. Invertibility of A^TA forces $x = 0$, so the columns of A are independent. There is no step requiring A itself to be square or invertible.

Exercise 6.5.21: The converse and its rank consequences

(a) Independent columns give $\operatorname{Nul} A = \{0\}$. Exercise 19 makes $\operatorname{Nul}(A^TA) = \{0\}$ too. Since A^TA is square $n \times n$, the Invertible Matrix Theorem makes it invertible.

(b) There are n independent columns in $\mathbb{R}^m$, so $n \leq m$. Equivalently, n pivot columns require at least n rows.

(c) All n columns are pivot columns, so $\operatorname{rank} A = n$.

Exercise 6.5.22: The Gram matrix has the same rank

Both A and A^TA have n columns, so rank-nullity gives $\operatorname{rank} A = n - \dim \operatorname{Nul} A$ and $\operatorname{rank}(A^TA) = n - \dim \operatorname{Nul}(A^TA)$. The null spaces are equal by Exercise 19, so the subtracted dimensions are equal. Thus $\operatorname{rank}(A^TA) = \operatorname{rank} A$, even when the columns are dependent.

Exercise 6.5.23: Projection without an orthogonal column basis

Independent columns make A^TA invertible. Solving the normal equations gives $\widehat{x} = (A^TA)^{-1}A^Tb$. Multiply by A to get the fitted output:

$$\boxed{\widehat{b} = A(A^TA)^{-1}A^Tb}.$$

This is a formula for the projection in $\mathbb{R}^m$, not just its coefficient vector in $\mathbb{R}^n$. Orthogonality follows by computing $A^T(b - \widehat{b}) = A^Tb - A^TA(A^TA)^{-1}A^Tb = 0$.

Exercise 6.5.24: Orthonormal columns simplify the normal equations

If the columns are orthonormal, their Gram matrix is $A^TA = I_n$. Thus the normal equations reduce immediately to $\boxed{\widehat{x} = A^Tb}$. The corresponding projection is AA^Tb. Full column rank is automatic because no nontrivial combination of orthonormal columns can be zero.

Exercise 6.5.25: Fit two incompatible equations with a common left side

Let $s = x + y$. The squared error is

$$(2 - s)^2 + (4 - s)^2 = 2(s - 3)^2 + 2.$$

This is minimized exactly when $s = 3$. Hence all least-squares solutions are $(x, y) = (t, 3 - t)$, $t \in \mathbb{R}$. They all produce the projected output $(3, 3)^T$ and error $\sqrt{2}$. The fitted output is unique, while the two input weights are not.

Exercise 6.5.26: Design the approximate low-pass filter from the actual table

Use the table's rounded value $q = 0.7$, and define the periodic lists

$$y = (1, q, 0, -q, -1, -q, 0, q), \qquad w = (1, -q, 0, q, -1, q, 0, -q).$$

Indices are reduced modulo 8. Form an eight-row block $Y_k = (y_{k+2}, y_{k+1}, y_k)$ and another block $Z_k = (w_{k+2}, w_{k+1}, w_k)$ for $k = 0, \dots, 7$. Then the requested 16×3 system is explicitly

$$A = \begin{bmatrix} Y \\ Z \end{bmatrix}, \quad x = (a_0, a_1, a_2)^T, \quad b = (q, 0, -q, -1, -q, 0, q, 1, 0, 0, 0, 0, 0, 0, 0, 0)^T.$$

For clarity the two blocks are

$$Y = \begin{bmatrix} 0 & q & 1 \\ -q & 0 & q \\ -1 & -q & 0 \\ -q & -1 & -q \\ 0 & -q & -1 \\ q & 0 & -q \\ 1 & q & 0 \\ q & 1 & q \end{bmatrix}, \quad Z = \begin{bmatrix} 0 & -q & 1 \\ q & 0 & -q \\ -1 & q & 0 \\ q & -1 & q \\ 0 & q & -1 \\ -q & 0 & q \\ 1 & -q & 0 \\ -q & 1 & -q \end{bmatrix}.$$

Column dot products give $A^T A = (198/25) I_3$ and $A^T b = (14/5, 99/25, 14/5)^T$. The diagonal normal equations therefore yield

$$\boxed{a_0 = a_2 = 35/99 \approx 0.35353535, \qquad a_1 = 1/2}.$$

The normal-equation residual is exactly zero with these coefficients and the rational table entry $q = 7/10$. More generally the same computation for a table value q gives $a_0 = a_2 = q/(1 + 2q^2)$ and $a_1 = 1/2$. Putting $q = \sqrt{2}/2$ gives $a_0 = a_2 = \sqrt{2}/4$; using $q = 0.707$ gives a very close approximation. The rounded-table answer should not be silently replaced by the exact-trigonometric answer.

6.6 Applications to linear models

A model is linear in its *unknown parameters*, even if it contains x^2, $\sin x$, or $e^{-0.02x}$. For observations (x_i, y_i), write the predicted value as a row of a design matrix X times a parameter vector β. The observation equation is $y = X\beta + \varepsilon$. Least squares minimizes $\|\varepsilon\|$, and the fitted parameters solve $X^T X\widehat{\beta} = X^T y$. As a check, the fitted residual always satisfies $X^T(y - X\widehat{\beta}) = 0$. We retain unrounded coefficients in computations and round only displayed predictions.

Practice problem 6.6.P1: A seasonal term is still linear in its coefficient

For the proposed seasonal model, use

$$X = \begin{bmatrix} 1 & x_1 & \sin(2\pi x_1/12) \\ \vdots & \vdots & \vdots \\ 1 & x_n & \sin(2\pi x_n/12) \end{bmatrix}, \quad \beta = (\beta_0, \beta_1, \beta_2)^T, \quad y = (y_1, \ldots, y_n)^T.$$

The ith product row equals the predicted sales at month x_i. The sine values are known once the observation times are supplied; only their coefficients are unknown, so this is a linear least-squares model.

Exercise 6.6.1: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{9}{10} \\ \frac{2}{5} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{9}{10} + (\frac{2}{5})x}.$$

The residual $\begin{bmatrix} \frac{1}{10} \\ -\frac{3}{10} \\ \frac{3}{10} \\ -\frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{5}$.

Exercise 6.6.2: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 12 \\ 12 & 46 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 25 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} -\frac{3}{5} \\ \frac{7}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = -\frac{3}{5} + (\frac{7}{10})x}.$$

The residual $\begin{bmatrix} -\frac{1}{10} \\ \frac{1}{5} \\ -\frac{1}{5} \\ \frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{10}$.

Exercise 6.6.3: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 4 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{11}{10} \\ \frac{13}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{11}{10} + (\frac{13}{10})x}.$$

The residual $\begin{bmatrix} \frac{1}{5} \\ -\frac{1}{10} \\ -\frac{2}{5} \\ \frac{3}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{3}{10}$.

Exercise 6.6.4: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}, \qquad y = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 16 \\ 16 & 74 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{43}{10} \\ -\frac{7}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{43}{10} + (-\frac{7}{10})x}.$$

The residual $\begin{bmatrix} \frac{1}{10} \\ -\frac{1}{5} \\ \frac{1}{5} \\ -\frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{10}$.

Exercise 6.6.5: Exactly when does a line fit have unique parameters?

By Section 6.5, the normal equations have a unique solution exactly when the two columns of X are independent. They are the all-ones column and $(x_1, \ldots, x_n)^T$. If all x_i equal the same c, the second column is c times the first, so independence and uniqueness fail. Conversely, suppose $ax_i + b = 0$ at all data points and two abscissas differ. Subtracting the equations at those two points forces $a = 0$, then $b = 0$. Thus the columns are independent precisely when at least two distinct x-coordinates occur. Repeated observations at one x cannot identify both an intercept and a slope uniquely.

Exercise 6.6.6: Three distinct abscissas identify a quadratic fit

The three design columns are $1, x, x^2$. A dependence among them would give a polynomial $c_0+c_1t+c_2t^2$ that vanishes at every observed abscissa, in particular at three distinct ones. A nonzero polynomial of degree at most two cannot have three distinct roots. Hence all coefficients must be zero, proving independence of the columns. The full-column-rank least-squares theorem then gives a unique best-fitting parameter vector. The fitted polynomial may happen to have zero quadratic coefficient; uniqueness concerns the polynomial, not a requirement that its degree be exactly two.

Exercise 6.6.7: Fit a quadratic with no constant term

(a) Put the known powers into the design columns.

For the five observations, the model is

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}, \quad y = (1.8, 2.7, 3.4, 3.8, 3.9)^T, \quad \beta = (\beta_1, \beta_2)^T.$$

There is no column of ones, because the proposed model has no intercept.

(b) Solve and state the curve.

The exact normal equations are

$$\begin{bmatrix} 55 & 225 \\ 225 & 979 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 521/10 \\ 403/2 \end{bmatrix}.$$

Their solution is $\widehat{\beta} = (14171/8050, -32/161)^T$. Thus the least-squares curve is

$$y = \frac{14171}{8050}x - \frac{32}{161}x^2 \approx 1.760373x - 0.198758x^2.$$

Substitution in the normal equations verifies both residual moments are zero. This certifies the fit with the supplied decimal data treated exactly.

Exercise 6.6.8: Fit a cubic cost model without fixed costs

(a) Record the general model.

Use $X_{i,:} = (x_i, x_i^2, x_i^3)$, $y = (y_1, \ldots, y_n)^T$, and $\beta = (\beta_1, \beta_2, \beta_3)^T$. Again no column of ones appears. The residual equation is $y = X\beta + \varepsilon$.

(b) Use the eight supplied observations.

Here $x_i = 4, 6, 8, 10, 12, 14, 16, 18$, and the observed costs are $1.58, 2.08, 2.50, 2.80, 3.10, 3.40, 3.80, 4.32$, in the stated thousand-unit scales. The normal equations are

$$\begin{bmatrix} 1136 & 16192 & 245312 \\ 16192 & 245312 & 3866368 \\ 245312 & 3866368 & 62617856 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 7254/25 \\ 100636/25 \\ 1501896/25 \end{bmatrix}.$$

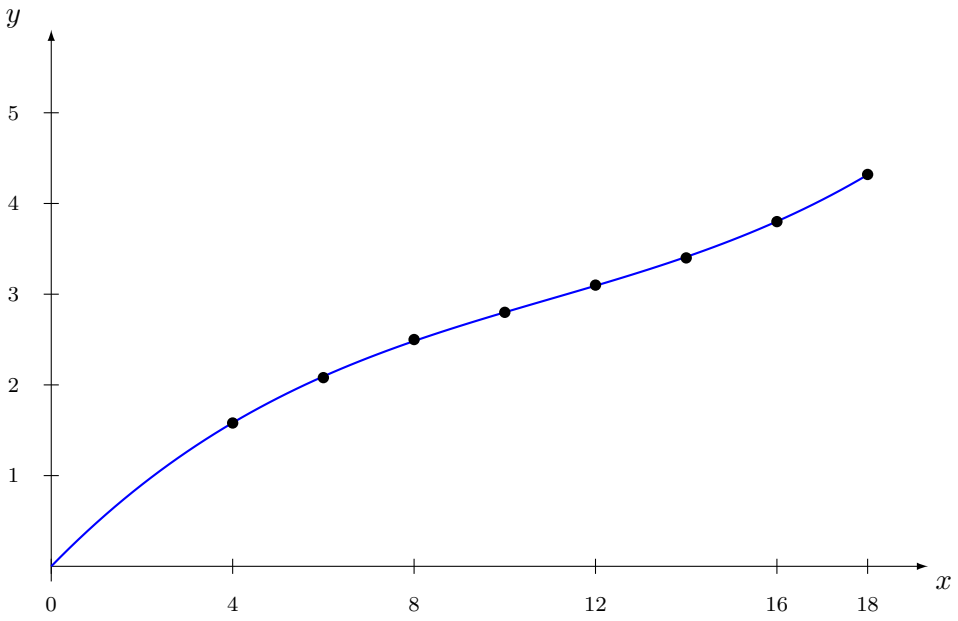
Solving with exact rational data gives

$$\widehat{\beta} = (8970503/17479000,\ -319181/9534000,\ 15221/14982000)^T.$$

The fitted curve is therefore

$$y \approx 0.513216031x - 0.0334781833x^2 + 0.00101595248x^3.$$

The exact residual satisfies $X^T(y - X\widehat{\beta}) = 0$. The graph below shows the fitted curve in blue and all eight observations as black points; both axes use the problem's thousand-unit scale.



Exercise 6.6.9: A trigonometric model linear in two coefficients

Use

$$X = \begin{bmatrix} \cos 1 & \sin 1 \\ \cos 2 & \sin 2 \\ \cos 3 & \sin 3 \end{bmatrix}, \quad y = (7.9, 5.4, -0.9)^T, \quad \beta = (A, B)^T.$$

Angles are in radians. Although the basis functions are trigonometric, the unknowns A, B enter linearly. Minimize $\|y - X\beta\|$ by solving $X^T X\widehat{\beta} = X^T y$, or use QR on X directly. A numerical solution gives $A \approx 2.34212062$, $B \approx 7.44750155$, so $y \approx 2.34212062\cos x + 7.44750155\sin x$. Using unrounded coefficients, the normal residual is below 3×10^{-14} in absolute value, a check on the computed fit.

Exercise 6.6.10: Estimate two initial radioactive masses

(a) Treat the decay functions as known columns.

At times $t_i = 10, 11, 12, 14, 15$, use

$$X_{i,:} = (e^{-0.02t_i}, e^{-0.07t_i}), \quad \beta = (M_A, M_B)^T,$$

with observation vector $(21.34, 20.68, 20.05, 18.87, 18.30)^T$. The decay constants are supplied, so the two masses, not the exponent rates, are the unknown linear parameters.

(b) Fit the two masses.

Numerical least squares gives $M_A \approx 19.94109495$ and $M_B \approx 10.10151413$ grams. Thus the fitted curve is

$$y(t) \approx 19.94109495e^{-0.02t} + 10.10151413e^{-0.07t}.$$

Both fitted masses are positive. The residual norm is approximately 0.0114745 grams, and the normal residual computed from unrounded coefficients is below 3×10^{-14}. The residual is not assumed to vanish, because the measured data are rounded.

Exercise 6.6.11: Linearize the comet equation in the correct parameters

The relation $r = \beta + e(r\cos\theta)$ is linear in β, e. For the given observations, use design rows $(1, r_i\cos\theta_i)$, not $(1, \cos\theta_i)$, and observation vector (r_i). Least squares gives $\beta \approx 1.45092922$ and $e \approx 0.81105251$; the fitted normal residual is below 2×10^{-14}. Since $0 < e < 1$, the predicted orbit is an ellipse. For a new angle, solve the original model for its new radius:

$$r = \frac{\beta}{1 - e\cos\theta}, \qquad r(4.6) \approx 1.32995445.$$

The denominator is positive here (indeed $e < 1$ makes it positive at every angle).

Conclusion. An elliptic orbit, with predicted radius about 1.330 in the same units as the observed radii.

Exercise 6.6.12: Regress pressure on logarithmic weight

Use design rows $(1, \ln w_i)$ and observation vector $(91, 98, 103, 110, 112)^T$. If the printed rounded logarithms $3.78, 4.11, 4.39, 4.73, 4.88$ are used, the fitted line is

$$p \approx 18.56421016 + 19.24070120\ln w.$$

At $w = 100$, using $\ln 100 \approx 4.605170186$, this predicts $p \approx 107.17091367$ mmHg. For clarity about precision: recomputing the logarithms directly from $w = 44, 61, 81, 113, 131$ instead of using their rounded table entries gives $p \approx 17.92434012 + 19.38499925\ln w$, and a prediction 107.19556074 mmHg. Both computations satisfy their respective normal equations to roundoff.

Conclusion. About 107 mmHg. The small difference between the two more detailed predictions comes from rounding the predictor data.

Exercise 6.6.13: Fit takeoff position and differentiate the fitted polynomial

(a) Build the cubic design matrix from the thirteen times.

For $t = 0, 1, \ldots, 12$, take row $(1, t, t^2, t^3)$ and the given position vector in feet. The normal equations have coefficient matrix

$$X^TX = \begin{bmatrix} 13 & 78 & 650 & 6084 \\ 78 & 650 & 6084 & 60710 \\ 650 & 6084 & 60710 & 630708 \\ 6084 & 60710 & 630708 & 6735950 \end{bmatrix},$$

with right side $X^Ty = (19001/5, 351277/10, 3480639/10, 35998009/10)^T$. Solving gives the exact coefficient vector

$$\widehat{\beta} = (-89/104,\ 75315/16016,\ 222437/40040,\ -313/11440)^T.$$

Thus, approximately,

$$y(t) = -0.855769231 + 4.702485015t + 5.555369630t^2 - 0.02736013986t^3.$$

The exact coefficients make $X^T(y - X\widehat{\beta}) = 0$; distinct times ensure the four design columns are independent, so the fit is unique.

(b) Differentiate before evaluating.

The fitted velocity is $y'(t) = \widehat{\beta}_1 + 2\widehat{\beta}_2 t + 3\widehat{\beta}_3 t^2$. At $t = 4.5$ this gives $\boxed{53.03868319 \text{ feet/second}}$. This estimates instantaneous velocity from the fitted curve, rather than dividing the fitted position by the elapsed time.

Exercise 6.6.14: The fitted line passes through the mean point

The residual $r = y - X\widehat{\beta}$ is orthogonal to each column of X, including the all-ones column. Thus its entries sum to zero: $\sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0$. Divide by $n > 0$ and use the definitions of the means to obtain

$$\overline{y} = \widehat{\beta}_0 + \widehat{\beta}_1 \overline{x}.$$

This says exactly that $(\overline{x}, \overline{y})$ lies on the fitted line.

Exercise 6.6.15: Derive the scalar normal equations

For design columns $\mathbf{1}$ and x, the four Gram entries are $\mathbf{1}^T\mathbf{1} = n$, $\mathbf{1}^T x = x^T\mathbf{1} = \sum x_i$, and $x^T x = \sum x_i^2$. Similarly, $X^T y = (\sum y_i, \sum x_i y_i)^T$. Multiplication of the resulting two-by-two system gives

$$n\widehat{\beta}_0 + \widehat{\beta}_1 \sum x_i = \sum y_i, \qquad \widehat{\beta}_0 \sum x_i + \widehat{\beta}_1 \sum x_i^2 = \sum x_i y_i,$$

which are precisely equations (7).

Exercise 6.6.16: Invert the two-by-two normal matrix

Write $S_x = \sum x_i$, $S_{xx} = \sum x_i^2$, $S_y = \sum y_i$, and $S_{xy} = \sum x_i y_i$. When at least two abscissas differ, $D = nS_{xx} - S_x^2 = n\sum_i (x_i - \overline{x})^2 > 0$. Thus the normal matrix inverse is

$$\frac{1}{D}\begin{bmatrix} S_{xx} & -S_x \\ -S_x & n \end{bmatrix}.$$

Multiplying it by $(S_y, S_{xy})^T$ gives

$$\boxed{\widehat{\beta}_0 = \frac{S_{xx}S_y - S_x S_{xy}}{D}, \qquad \widehat{\beta}_1 = \frac{nS_{xy} - S_x S_y}{D}}.$$

Equivalently $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$. If all abscissas coincide, $D = 0$ and this inverse formula is inapplicable, consistent with the failure of unique parameter identification.

Exercise 6.6.17: Centering the predictor makes the columns orthogonal

(a) Rewrite the actual data from Example 1.

Their mean abscissa is $(2+5+7+8)/4 = 5.5$. The transformed data are

$$(-3.5, 1), \quad (-0.5, 2), \quad (1.5, 3), \quad (2.5, 3).$$

The design matrix has columns $\mathbf{1}$ and $x^* = (-3.5, -0.5, 1.5, 2.5)^T$. Their dot product is the sum of x^*, which is zero, proving orthogonality.

(b) Solve the diagonal normal equations.

Here $X^TX = \operatorname{diag}(4, 21)$ and $X^Ty = (9, 7.5)^T$. Hence $\widehat{\beta}_0 = 9/4$ and $\widehat{\beta}_1 = 5/14$.

Conclusion. The centered fitted line is $y = 9/4 + (5/14)x^*$, with $x^* = x - 5.5$. Its uncentered form is $y = 2/7 + (5/14)x$.

Exercise 6.6.18: A general centered design has a diagonal Gram matrix

The two off-diagonal Gram entries are both $\sum_i x_i$, which is zero by the centered-coordinate assumption. Therefore

$$X^TX = \begin{bmatrix} n & 0 \\ 0 & \sum_i x_i^2 \end{bmatrix}.$$

If every centered abscissa is zero, the second diagonal entry vanishes; diagonal does not by itself mean invertible.

Exercise 6.6.19: Use an orthogonal decomposition of the observed vector

Write $y = X\widehat{\beta} + r$. The normal equations give $X^Tr = 0$, so r is orthogonal to $\operatorname{Col} X$, which contains $X\widehat{\beta}$. The Pythagorean theorem therefore applies:

$$\|y\|^2 = \|X\widehat{\beta}\|^2 + \|r\|^2.$$

With the problem's notation, this is $\mathrm{SS}(T) = \mathrm{SS}(R) + \mathrm{SS}(E)$. The zero-mean convention lets the total be interpreted as centered variation; the displayed norm identity itself follows from orthogonality.

Exercise 6.6.20: A computational formula for the fitted and residual sums

Using the normal equations in the middle step,

$$\|X\widehat{\beta}\|^2 = \widehat{\beta}^TX^TX\widehat{\beta} = \widehat{\beta}^TX^Ty.$$

Subtract this from the Pythagorean identity in Exercise 19 to obtain

$$\boxed{\mathrm{SS}(E) = y^Ty - \widehat{\beta}^TX^Ty}.$$

This avoids explicitly summing every residual square once the fit has been computed. In finite-precision arithmetic direct residual norms can be more stable when this subtraction nearly cancels two large numbers.

6.7 Inner product spaces

An inner product assigns real numbers $\langle u, v\rangle$ and satisfies symmetry, additivity, homogeneity, and positive definiteness. The associated norm is $\|u\| = \sqrt{\langle u, u\rangle}$; orthogonality means $\langle u, v\rangle = 0$. The formula depends on the space: a weighted coordinate product, evaluation of polynomials, and integration of functions are different inner products. For a nonzero vector p, projection onto its span is $\langle q, p\rangle p/\langle p, p\rangle$. For an orthogonal list, add these separate projections. Always use the inner product stipulated in the exercise rather than substituting the ordinary coordinate dot product.

Practice problem 6.7.P1: A zero vector has zero inner product with every vector

By homogeneity, $\langle 0, v\rangle = \langle 0u, v\rangle = 0\langle u, v\rangle = 0$, for any u in the space. Symmetry then gives $\langle v, 0\rangle = \langle 0, v\rangle = 0$. The choice of u does not matter, because its scalar multiple is zero.

Practice problem 6.7.P2: Additivity in the second argument

Use symmetry to move the sum into the first argument, where additivity is an axiom:

$$\langle u, v + w\rangle = \langle v + w, u\rangle = \langle v, u\rangle + \langle w, u\rangle = \langle u, v\rangle + \langle u, w\rangle.$$

The last equality applies symmetry to each term separately.

Exercise 6.7.1: A weighted norm and its orthogonal line

Example 1 uses $\langle u, v\rangle = 4u_1v_1 + 5u_2v_2$.

(a) Apply those weights to the stated vectors.

For $x = (1, 1)$ and $y = (5, -1)$, $\|x\|^2 = 4 + 5 = 9$, $\|y\|^2 = 4(25) + 5 = 105$, and $\langle x, y\rangle = 20 - 5 = 15$. Thus $\|x\| = 3$, $\|y\| = \sqrt{105}$, and $|\langle x, y\rangle|^2 = 225$.

(b) Solve the orthogonality equation.

A vector $z = (z_1, z_2)$ is orthogonal to y exactly when $20z_1 - 5z_2 = 0$, or $z_2 = 4z_1$.

Conclusion. All such vectors are $t(1, 4)$, $t \in \mathbb{R}$, including zero.

Exercise 6.7.2: Verify Cauchy–Schwarz with the weighted product

For $x = (3, -2)$ and $y = (-2, 1)$,

$$\langle x, y\rangle = 4(3)(-2) + 5(-2)(1) = -34,$$

$$\|x\|^2 = 4(9) + 5(4) = 56, \qquad \|y\|^2 = 4(4) + 5(1) = 21.$$

Therefore $|\langle x, y\rangle|^2 = 1156 \le 1176 = \|x\|^2\|y\|^2$. Both sides of $|\langle x, y\rangle| \le \|x\|\|y\|$ are nonnegative, so the squared comparison proves the desired inequality.

Evaluation convention for Exercises 3–8. The inner product is the dot product of the three value lists at $-1, 0, 1$, not of the polynomial coefficient lists. For Exercise 3, p and q have value vectors $(3, 4, 5)$ and $(1, 5, 1)$. For Exercise 4, their value vectors are $(-4, 0, 2)$ and $(5, 3, 5)$.

Exercise 6.7.3: Evaluate before multiplying

For $p(t) = 4 + t$ and $q(t) = 5 - 4t^2$,

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1) = 3(1) + 4(5) + 5(1) = \boxed{28}.$$

Each product uses values at the same input.

Exercise 6.7.4: A negative inner product is allowed

For $p(t) = 3t - t^2$ and $q(t) = 3 + 2t^2$,

$$\langle p, q\rangle = (-4)(5) + 0(3) + 2(5) = \boxed{-10}.$$

Only self-products must be nonnegative; products of different vectors can be negative, as this example shows.

Exercise 6.7.5: Norms from the first pair of value lists

Using the value vectors $(3, 4, 5)$ and $(1, 5, 1)$,

$$\|p\| = \sqrt{9 + 16 + 25} = \boxed{5\sqrt{2}}, \qquad \|q\| = \sqrt{1 + 25 + 1} = \boxed{3\sqrt{3}}.$$

The norm is the square root of the sum of squared values, not its square.

Exercise 6.7.6: Norms from the second pair

The value vectors are $(-4, 0, 2)$ and $(5, 3, 5)$, so

$$\|p\| = \sqrt{16 + 0 + 4} = \boxed{2\sqrt{5}}, \qquad \|q\| = \sqrt{25 + 9 + 25} = \boxed{\sqrt{59}}.$$

These are norms in the specified evaluation inner product.

Exercise 6.7.7: Project onto the first polynomial line

For the pair in Exercise 3, $\langle q, p\rangle = 28$ and $\langle p, p\rangle = 50$. Since $p \neq 0$, projection onto $\mathrm{Span}\{p\}$ is

$$\mathrm{proj}_{\mathrm{Span}\{p\}}\, q = \frac{28}{50}p = \boxed{\frac{14}{25}(4 + t)}.$$

A check is $\langle q - (14/25)p, p\rangle = 28 - (14/25)50 = 0$. Thus the residual is orthogonal to the whole one-dimensional target space.

Exercise 6.7.8: Project onto the second polynomial line

For the pair in Exercise 4, the coefficient is $\langle q, p\rangle/\langle p, p\rangle = -10/20 = -1/2$. Hence the projection is $\boxed{-\frac{1}{2}(3t - t^2) = (t^2 - 3t)/2}$. Its residual check is $-10 - (-1/2)20 = 0$. A negative projection coefficient is perfectly possible.

Exercise 6.7.9: Remove the lower-degree part of a quadratic

Use evaluation nodes $-3, -1, 1, 3$. The value lists for p_0, p_1, p_2 are $(1, 1, 1, 1)$, $(-3, -1, 1, 3)$, and $(9, 1, 1, 9)$.

(a) Project onto the first two orthogonal polynomials.

The first two lists have dot product zero. Also $\langle p_2, p_0\rangle = 20$, $\langle p_0, p_0\rangle = 4$, and $\langle p_2, p_1\rangle = -27 - 1 + 1 + 27 = 0$. Thus the projection is $5p_0 = 5$.

(b) Form and scale the orthogonal remainder.

The remainder $t^2 - 5$ has values $(4, -4, -4, 4)$. Divide by four to get $\boxed{q(t) = (t^2 - 5)/4}$, whose values are $(1, -1, -1, 1)$. It is orthogonal to both p_0, p_1 and is nonzero. The three polynomials have distinct increasing degrees, so they are independent and span the same space as $1, t, t^2$.

Exercise 6.7.10: Best quadratic approximation to an odd cubic

In the same four-node inner product, t^3 is orthogonal to the even polynomials p_0 and q: the products cancel at opposite nodes. Its only nonzero coefficient in the orthogonal list is on $p_1 = t$:

$$\frac{\langle t^3, t\rangle}{\langle t, t\rangle} = \frac{81 + 1 + 1 + 81}{9 + 1 + 1 + 9} = \frac{41}{5}.$$

Conclusion. The best approximation is $(41/5)t$. The best member of a quadratic-polynomial space need not have a nonzero quadratic term.

Exercise 6.7.11: A cubic projected onto a five-node quadratic space

Here evaluation is at $-2, -1, 0, 1, 2$, and the source's orthogonal polynomials are $p_0 = 1$, $p_1 = t$, $p_2 = t^2 - 2$. Odd-even cancellation makes the projection coefficients on p_0, p_2 zero. The coefficient on t is

$$\frac{\sum t_i^4}{\sum t_i^2} = \frac{16 + 1 + 0 + 1 + 16}{4 + 1 + 0 + 1 + 4} = \frac{17}{5}.$$

Conclusion. The orthogonal projection is $(17/5)t$.

Exercise 6.7.12: Choose a scaled orthogonal cubic

Subtract the projection from Exercise 11 to obtain $r(t) = t^3 - (17/5)t$. Its values at the five nodes are $(-6/5, 12/5, 0, -12/5, 6/5)$. Multiplying by $5/6$ yields

$$\boxed{p_3(t) = (5t^3 - 17t)/6},$$

with the requested values $(-1, 2, 0, -2, 1)$. The remainder is orthogonal to all of $\text{Span}\{p_0, p_1, p_2\}$. Its degree is three, so it is nonzero and adds a new direction. Thus the four polynomials are an orthogonal basis for $\mathcal{P}_3$.

Exercise 6.7.13: Transport the Euclidean product through an invertible map

Define $\langle u, v\rangle = (Au)^T(Av)$. Symmetry follows from symmetry of the dot product. For addition, $A(u+w) = Au + Aw$, so $\langle u+w, v\rangle = (Au+Aw)^T Av = \langle u, v\rangle + \langle w, v\rangle$. Similarly $A(cu) = cAu$ gives homogeneity. Finally $\langle u, u\rangle = \|Au\|^2 \geq 0$. If it is zero, $Au = 0$; invertibility gives $u = 0$. Conversely $u = 0$ clearly gives zero. All four axioms hold. The last step is why invertibility, rather than merely linearity of A, was required.

Exercise 6.7.14: The same construction for an injective transformation

Define $\langle u, v\rangle = T(u) \cdot T(v)$. Symmetry and the two linearity axioms follow exactly by applying linearity of T and the dot-product rules in $\mathbb{R}^n$. The self-product is $\|T(u)\|^2$, nonnegative and zero exactly when $T(u) = 0$. Because T is one-to-one and $T(0) = 0$, this happens only for $u = 0$. Thus positive definiteness holds, proving an inner product on V. Without injectivity, a nonzero kernel vector would have self-product zero and violate that axiom.

Exercise 6.7.15: Move a scalar out of the second argument

Symmetry and homogeneity in the first argument give

$$\langle u, cv\rangle = \langle cv, u\rangle = c\langle v, u\rangle = c\langle u, v\rangle.$$

These are real inner products, so no complex-conjugation convention is involved.

Exercise 6.7.16: The distance between two orthonormal vectors

Expand using bilinearity and symmetry:

$$\|u - v\|^2 = \langle u, u\rangle - 2\langle u, v\rangle + \langle v, v\rangle = 1 - 0 + 1 = 2.$$

Taking the nonnegative square root gives $\boxed{\|u - v\| = \sqrt{2}}$.

Exercise 6.7.17: Recover an inner product from two norms

Expand the two squared norms: $\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$ and $\|u-v\|^2 = \|u\|^2 - 2\langle u, v\rangle + \|v\|^2$. Subtracting cancels both self-products and leaves $\|u+v\|^2 - \|u-v\|^2 = 4\langle u, v\rangle$. Division by four proves the stated identity.

Exercise 6.7.18: The parallelogram identity

Add, instead of subtracting, the two expansions in Exercise 17. The mixed terms $2\langle u, v\rangle$ and $-2\langle u, v\rangle$ cancel, leaving

$$\boxed{\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2}.$$

The proof uses only inner-product axioms, so it is not restricted to coordinate vectors.

Exercise 6.7.19: Arithmetic mean dominates geometric mean

Use the ordinary dot product on the two supplied coordinate vectors. Their dot product is $2\sqrt{ab}$, while each squared norm is $a+b$. Cauchy–Schwarz gives $2\sqrt{ab} \le \sqrt{a+b}\sqrt{a+b} = a+b$, where nonnegativity of a, b makes the square roots defined and the left side nonnegative. Divide by two: $\boxed{\sqrt{ab} \le (a+b)/2}$. The case $a = b = 0$ is included.

Exercise 6.7.20: The square of a mean versus the mean of squares

For $u = (a, b)^T$ and $v = (1, 1)^T$, squared Cauchy–Schwarz gives $(a+b)^2 \le (a^2+b^2)(1+1) = 2(a^2+b^2)$. Divide by four to obtain

$$\boxed{\left(\frac{a+b}{2}\right)^2 \le \frac{a^2+b^2}{2}}.$$

This holds for all real a, b, with no nonnegativity assumption needed.

Exercise 6.7.21: An integral inner product that vanishes

The product expands as $(1-3t^2)(t-t^3) = t - 4t^3 + 3t^5$. Therefore

$$\langle f, g\rangle = \int_0^1 (t - 4t^3 + 3t^5)\,dt = \frac{1}{2} - 1 + \frac{1}{2} = \boxed{0}.$$

Both functions are continuous, so this is the specified inner product; they are orthogonal even though neither is zero.

Exercise 6.7.22: Another orthogonal pair of functions

Expanding $(5t-3)(t^3-t^2)$ gives $5t^4 - 8t^3 + 3t^2$. Thus $\langle f, g\rangle = \int_0^1 (5t^4 - 8t^3 + 3t^2)\,dt = 1 - 2 + 1 = \boxed{0}$. The integral of a product can cancel even when the functions have nonzero norms.

Exercise 6.7.23: An integral norm is a square root

For $f(t) = 1 - 3t^2$,

$$\|f\|^2 = \int_0^1 (1 - 6t^2 + 9t^4)\,dt = 1 - 2 + \frac{9}{5} = \frac{4}{5}.$$

Hence $\boxed{\|f\| = 2/\sqrt{5}}$. Nonnegativity determines the choice of square root.

Exercise 6.7.24: The norm of a cubic-minus-quadratic function

For $g(t) = t^3 - t^2$,

$$\|g\|^2 = \int_0^1 (t^6 - 2t^5 + t^4)\,dt = \frac{1}{7} - \frac{1}{3} + \frac{1}{5} = \frac{1}{105}.$$

Thus $\boxed{\|g\| = 1/\sqrt{105}}$.

Exercise 6.7.25: An orthogonal polynomial basis on a symmetric interval

On $[-1,1]$, oddness gives $\langle 1,t\rangle = 0$ and $\langle t,t^2\rangle = 0$. The projection of t^2 onto 1 has coefficient $(\int_{-1}^{1} t^2)/(\int_{-1}^{1} 1) = (2/3)/2 = 1/3$. Thus $t^2 - 1/3$ is orthogonal to both lower-degree polynomials. A conventional scaling gives the basis

$$\boxed{\{1,\ t,\ (3t^2-1)/2\}}.$$

The last polynomial has integral zero and its product with t is odd. All three are nonzero of distinct degrees, so they form a basis of $\mathrm{Span}\{1,t,t^2\}$. Their squared norms are $2, 2/3, 2/5$, respectively.

Exercise 6.7.26: Repeat the orthogonalization on a wider interval

On $[-2,2]$, the first two polynomials are again orthogonal by oddness. The coefficient to subtract from t^2 is $(\int_{-2}^{2} t^2)/(\int_{-2}^{2} 1) = (16/3)/4 = 4/3$. Scaling the remainder by 3 gives $\boxed{\{1,t,3t^2-4\}}$. The last polynomial has integral $3(16/3) - 4(4) = 0$, and its product with t is odd. Distinct degrees give independence and the same polynomial span. All squared norms are positive; the last is $256/5$.

Exercise 6.7.27: Complete the five-node orthogonal polynomial basis

Use the evaluation nodes $-2,-1,0,1,2$ and start with $p_0 = 1, p_1 = t, p_2 = t^2 - 2$. They are orthogonal. For t^3, only the projection onto t is nonzero, with coefficient $17/5$ as computed in Exercise 11. Set $r_3 = t^3 - (17/5)t$. For t^4, oddness makes the projections onto p_1 and r_3 zero. The constant projection coefficient is $34/5$, while

$$\frac{\langle t^4, p_2\rangle}{\langle p_2, p_2\rangle} = \frac{(\sum t_i^6) - 2(\sum t_i^4)}{14} = \frac{130-68}{14} = \frac{31}{7}.$$

Hence the final remainder is

$$r_4 = t^4 - \frac{34}{5} - \frac{31}{7}(t^2-2) = t^4 - \frac{31}{7}t^2 + \frac{72}{35}.$$

One convenient rescaled basis is

$$\boxed{\{1,t,t^2-2,5t^3-17t,35t^4-155t^2+72\}}.$$

Their distinct degrees prove independence, and Gram–Schmidt preserves the original span $\mathcal{P}_4$. Exact dot products of their five-node value vectors are zero for every distinct pair, verifying orthogonality.

Exercise 6.7.28: Orthogonalize powers of cosine by integration

Start with $q_0 = 1$ and $q_1 = \cos t$: their integral product over $[0, 2\pi]$ is zero. Since $\int \cos^2 t = \pi$ and $\int 1 = 2\pi$, the projection of $\cos^2 t$ onto the constant is $1/2$. Its projection onto $\cos t$ is zero because $\int_0^{2\pi} \cos^3 t\, dt = 0$. Thus $q_2 = \cos^2 t - 1/2$. For $\cos^3 t$, the constant and q_2 projections vanish by cancellation over a half-period. The remaining coefficient is

$$\frac{\int_0^{2\pi} \cos^4 t\, dt}{\int_0^{2\pi} \cos^2 t\, dt} = \frac{3\pi/4}{\pi} = \frac{3}{4}.$$

Therefore $q_3 = \cos^3 t - (3/4)\cos t$, giving the orthogonal basis

$$\boxed{\{1, \cos t, \cos^2 t - 1/2, \cos^3 t - (3/4)\cos t\}}.$$

The squared norms are $2\pi, \pi, \pi/4, \pi/16$, all nonzero. Equivalently the last two are $(1/2)\cos 2t$ and $(1/4)\cos 3t$, so the basis spans the same four-dimensional space as the original list.

6.8 Applications of inner product spaces

Two conventions matter. The book uses W to multiply *residuals*, so weighted least squares minimizes $\|W(y - X\beta)\|^2$. Therefore its normal matrix contains $W^TW = W^2$ when W is diagonal. Do not replace this by W unless the supplied weights have first been square-rooted. For Fourier approximations, the inner product is the ordinary integral $\langle f, g\rangle = \int_0^{2\pi} fg$; the constant coefficient is $a_0/2$, with $a_0 = (1/\pi)\int f$, while $a_k = (1/\pi)\int f\cos kt$ and $b_k = (1/\pi)\int f\sin kt$. These differing denominators arise because $\|1\|^2 = 2\pi$ and $\|\sin kt\|^2 = \|\cos kt\|^2 = \pi$ for positive integer k.

Practice problem 6.8.P1: Check the three integral products

For $q_1 = 1$, $q_2 = t$, $q_3 = 3t^2 - 4$ on $[-2, 2]$,

$$\langle q_1, q_2\rangle = \int_{-2}^{2} t\,dt = 0,$$

$$\langle q_1, q_3\rangle = [t^3 - 4t]_{-2}^{2} = 0,$$

$$\langle q_2, q_3\rangle = \int_{-2}^{2} (3t^3 - 4t)\,dt = 0.$$

The first and third vanish also by oddness. All three polynomials are nonzero, so the pairwise zero products prove an orthogonal set.

Practice problem 6.8.P2: Read approximations from a finite trigonometric expression

The function is already $f(t) = 3 - 2\sin t + 5\sin 2t - 6\cos 2t$. The first-order space retains only the constant and frequency-one terms, so its approximation is $3 - 2\sin t$. The third-order space contains all the displayed terms (there are no frequency-three terms to add), so its approximation is f itself. Orthogonality makes the discarded higher-frequency terms perpendicular to the lower-order space; this is why simply retaining terms gives the least-squares approximation here.

Exercise 6.8.1: Half-size residual multipliers at the two endpoints

Use design rows $(1, x_i)$ for $x_i = -2, -1, 0, 1, 2$ and $y = (0, 0, 2, 4, 4)^T$. Following the book's convention, take $W = \mathrm{diag}(1, 2, 2, 2, 1)$: the endpoint residuals receive half the multiplier of the interior ones. Scaling all multipliers by a common positive factor does not change the fit. The weighted normal equations are

$$X^TW^2X\widehat{\beta} = X^TW^2y, \qquad \begin{bmatrix} 14 & 0 \\ 0 & 16 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 28 \\ 24 \end{bmatrix}.$$

Thus $\widehat{\beta}_0 = 2$ and $\widehat{\beta}_1 = 3/2$.

Conclusion. The fitted line is $y = 2 + (3/2)x$. Here "half" refers to the residual multiplier used in the text, whose square enters the objective.

Exercise 6.8.2: A common rescaling leaves the minimizers unchanged

The second weighting matrix is $W_2 = 2W_1$, because every point's multiplier is doubled. For any parameter vector,

$$\|W_2(y - X\beta)\|^2 = 4\|W_1(y - X\beta)\|^2.$$

Multiplication by the positive constant 4 preserves every ordering of objective values, so the minimizing parameter vectors are exactly the same. The numerical minimum changes by that factor, but the fitted curve does not. This also holds when the minimizer is not unique.

Exercise 6.8.3: Add the cubic trend to the quadratic trend

Use nodes $-2, -1, 0, 1, 2$ and observations $g = (3, 5, 5, 4, 3)$. The orthogonal polynomials from Example 2 and the question are $p_0 = 1, p_1 = t, p_2 = t^2 - 2, p_3 = (5t^3 - 17t)/6$. Their inner products with the data and squared norms are

j	0	1	2	3
$\langle g, p_j \rangle$	20	-1	-7	2
$\langle p_j, p_j \rangle$	5	10	14	10

The cubic values are $(-1, 2, 0, -2, 1)$, so its coefficient is $2/10 = 1/5$. The fitted cubic trend is

$$\widehat{g} = 4p_0 - \tfrac{1}{10}p_1 - \tfrac{1}{2}p_2 + \tfrac{1}{5}p_3 = \boxed{5 - \tfrac{2}{3}t - \tfrac{1}{2}t^2 + \tfrac{1}{6}t^3}.$$

Because the basis is orthogonal, adding the cubic term does not require recomputing the earlier three coefficients.

Exercise 6.8.4: Orthogonal trends at six evenly spaced nodes

(a) Check the evaluation vectors.

At $-5, -3, -1, 1, 3, 5$, the proposed polynomials have lists

$$p_0 : (1, 1, 1, 1, 1, 1), \quad p_1 : (-5, -3, -1, 1, 3, 5),$$

$$p_2 : (5, -1, -4, -4, -1, 5).$$

The lists for p_1 and p_2 each sum to zero, so both are orthogonal to p_0. Their mutual product sums to zero by odd-even cancellation. Their squared norms are $6, 70, 84$, all positive. Thus they form the required orthogonal basis of the quadratic space.

(b) Project the actual data onto that basis.

For $g = (1, 1, 4, 4, 6, 8)$, the three inner products are $24, 50, 6$. Hence

$$\widehat{g} = \frac{24}{6}p_0 + \frac{50}{70}p_1 + \frac{6}{84}p_2 = 4 + \frac{5}{7}t + \frac{1}{14}\left(\frac{3}{8}t^2 - \frac{35}{8}\right).$$

Equivalently the quadratic is $\boxed{\widehat{g}(t) = 59/16 + (5/7)t + (3/112)t^2}$. The projection residual is orthogonal to all three basis lists, verifying the fit.

Exercise 6.8.5: Sines of distinct integer frequencies are orthogonal

For unequal positive integers m, n, use $2\sin mt\sin nt = \cos((m-n)t) - \cos((m+n)t)$. Both $m-n$ and $m+n$ are nonzero integers. Their cosine integrals on $[0, 2\pi]$ are zero, since $\int_0^{2\pi}\cos(kt)\,dt = [\sin(kt)/k]_0^{2\pi} = 0$. Thus the integral of the original product is zero, proving orthogonality.

Exercise 6.8.6: Sine and cosine terms are orthogonal even at equal frequency

Use $2\sin mt\cos nt = \sin((m+n)t) + \sin((m-n)t)$. Every nonzero integer-frequency sine integrates to zero over $[0, 2\pi]$. If $m = n$, the second term is identically $\sin 0 = 0$, so it also has zero integral without any division by zero. Thus $\langle \sin mt, \cos nt\rangle = 0$ for all the stated positive integers.

Exercise 6.8.7: The Fourier basis elements have squared norm pi

For positive integer k, the identities $\cos^2 kt = (1+\cos 2kt)/2$ and $\sin^2 kt = (1-\cos 2kt)/2$ give

$$\|\cos kt\|^2 = \tfrac{1}{2}(2\pi) + 0 = \pi, \qquad \|\sin kt\|^2 = \tfrac{1}{2}(2\pi) - 0 = \pi.$$

The Fourier setting uses integer frequencies. For an arbitrary noninteger real k, the boundary term $\sin(4\pi k)/(4k)$ need not vanish, so the integer-frequency condition is what justifies these equalities.

Exercise 6.8.8: A constant shift changes only the constant coefficient

For $f(t) = t - 1$, the mean coefficient is $(1/(2\pi))\int_0^{2\pi}(t-1)\,dt = \pi - 1$. For $k \geq 1$, integration by parts gives $a_k = 0$ and $b_k = -2/k$; the constant -1 has zero product with those sines and cosines.

Conclusion. The third-order approximation is $\pi - 1 - 2\sin t - \sin 2t - (2/3)\sin 3t$.

Exercise 6.8.9: Reverse the slope of the linear function

For $f(t) = 2\pi - t$, the constant coefficient is π. The nonconstant Fourier coefficients of t have their signs reversed: $a_k = 0$, $b_k = 2/k$. These also follow directly by integrating $(2\pi - t)\sin kt$ by parts.

Conclusion. The third-order approximation is $\pi + 2\sin t + \sin 2t + (2/3)\sin 3t$.

Exercise 6.8.10: Project an integrable square wave onto continuous trigonometric functions

The square wave in the exercise has a jump at π, so it is not literally a member of $C[0, 2\pi]$. Nevertheless it is bounded and Riemann integrable, as are all its products with the finite continuous trigonometric basis. The finite-dimensional least-squares problem $\min_g \int_0^{2\pi}(f-g)^2$ is therefore well defined on that continuous trigonometric subspace. Its normal equations use the same integrals; no assertion about an inner product on all pointwise piecewise-continuous functions is needed. Values assigned at isolated jump endpoints do not change any of these integrals. The mean is zero. For $k \geq 1$,

$$a_k = \frac{1}{\pi}\left(\int_0^{\pi}\cos kt\,dt - \int_{\pi}^{2\pi}\cos kt\,dt\right) = 0,$$

$$b_k = \frac{1}{\pi}\left(\int_0^{\pi} \sin kt\,dt - \int_{\pi}^{2\pi} \sin kt\,dt\right) = \frac{2(1-(-1)^k)}{\pi k}.$$

Thus even sine coefficients vanish and odd ones equal $4/(\pi k)$.

Conclusion. The third-order approximation is $(4/\pi)(\sin t + (1/3)\sin 3t)$.

Exercise 6.8.11: An exact finite trigonometric identity

The identity $\sin^2 t = (1-\cos 2t)/2$ already expresses the function in the third-order target subspace. It is therefore its own best approximation, with zero error. Its only nonzero coefficients are the constant $1/2$ and the cosine-two coefficient $-1/2$.

Conclusion. $1/2 - (1/2)\cos 2t$. No integration is necessary.

Exercise 6.8.12: Use the cosine triple-angle identity

From $\cos 3t = 4\cos^3 t - 3\cos t$, solve for the cube: $\cos^3 t = (3/4)\cos t + (1/4)\cos 3t$. This lies entirely in the third-order target space, so the best approximation is exact, again with zero error.

Conclusion. $(3/4)\cos t + (1/4)\cos 3t$.

Exercise 6.8.13: Each coefficient depends linearly on the function

For example,

$$a_k(f+g) = \frac{1}{\pi}\int_0^{2\pi} (f+g)\cos kt = a_k(f) + a_k(g)$$

by linearity of the integral. Replacing cosine by sine proves the same identity for b_k, and taking the constant term proves it for a_0 too. The fixed denominators are the same for both functions, so no extra normalization appears when adding coefficients.

Exercise 6.8.14: One additional correct Fourier coefficient cannot worsen the fit

The difference is $h - g = b_2 \sin 2t$. Since b_2 is the Fourier coefficient of f, the residual $f - h$ is orthogonal to $\sin 2t$. Therefore the Pythagorean theorem gives

$$\|f-g\|^2 = \|(f-h) + b_2 \sin 2t\|^2 = \|f-h\|^2 + \pi b_2^2.$$

Conclusion. h is at least as close as g. It is strictly closer if $b_2 \neq 0$; if $b_2 = 0$, the two functions and their errors are identical.

Exercise 6.8.15: A weighted cubic takeoff fit

Use the thirteen times and position observations from Exercise 6.6.13, with design row $(1, t, t^2, t^3)$ and the stated diagonal residual multipliers $1, 1, 1, 0.9, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1$. The normal equations must use $X^T W^2 X$, not $X^T W X$. For reproducibility their coefficient matrix and right side

are

$$\begin{bmatrix} 6.66 & 22.23 & 120.77 & 797.19 \\ 22.23 & 120.77 & 797.19 & 5956.13 \\ 120.77 & 797.19 & 5956.13 & 48490.23 \\ 797.19 & 5956.13 & 48490.23 & 420477.17 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 747.844 \\ 4815.438 \\ 35420.468 \\ 285262.44 \end{bmatrix}.$$

Solving with the supplied decimals treated as exact rational data gives, after rounding for display,

$$y(t) \approx -0.2685118943 + 3.6095481671t + 5.8575839604t^2 - 0.04770303371t^3.$$

The exact fitted residual satisfies $X^T W^2 (y - X\widehat{\beta}) = 0$, which checks that this curve minimizes the weighted objective. Differentiate and evaluate at 4.5 seconds:

$$y'(4.5) = \widehat{\beta}_1 + 9\widehat{\beta}_2 + 60.75\widehat{\beta}_3 \approx \boxed{53.42984451 \text{ feet/second}}.$$

The later observations receive less influence because their residuals are multiplied by smaller numbers before squaring.

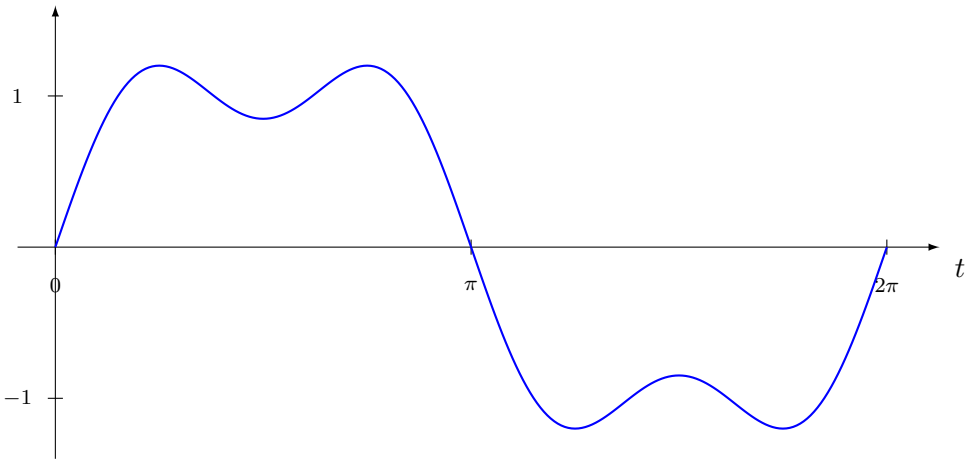
Exercise 6.8.16: Graph two Fourier orders and their periodic continuation

From Exercise 10, even sine coefficients vanish, so

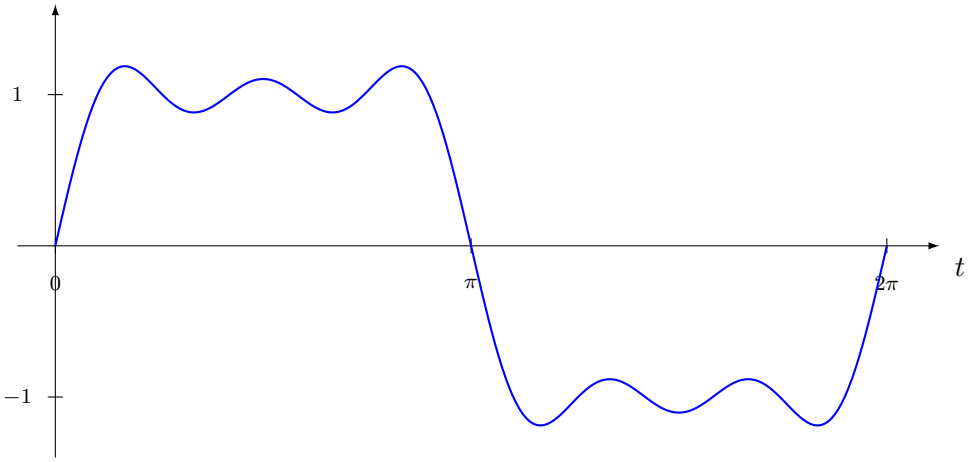
$$f_4(t) = \frac{4}{\pi}\left(\sin t + \frac{\sin 3t}{3}\right), \qquad f_5(t) = f_4(t) + \frac{4}{5\pi}\sin 5t.$$

Thus the fourth-order projection equals the third-order projection: its new frequency-four coefficients are zero. The fifth adds a nonzero frequency-five term. The requested three separate plots follow; the formulas determine the values exactly, while the curves are sampled densely for drawing. Both are 2π-periodic, explaining the continuation on the third plot.

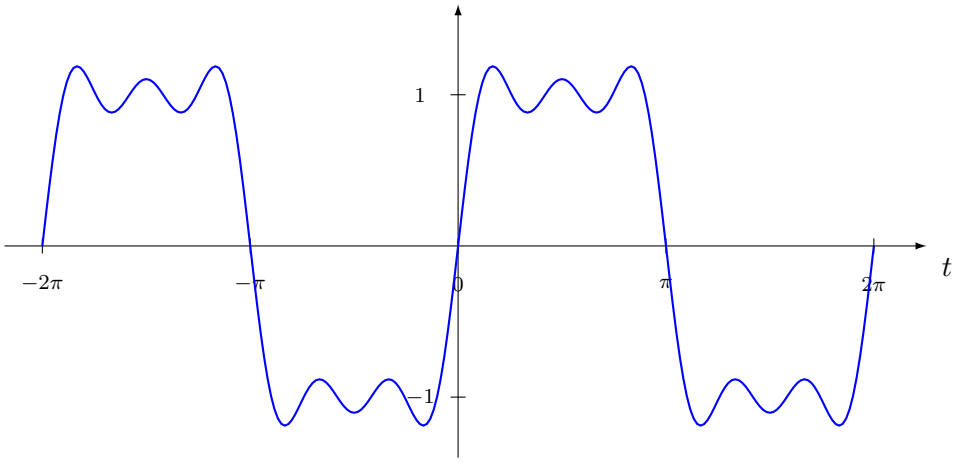
f_4 **on** $[0, 2\pi]$



f_5 **on** $[0, 2\pi]$



f_5 **on** $[-2\pi, 2\pi]$



Near the target's jumps the approximations overshoot; adding terms improves the integral fit without making these low-order continuous curves equal a discontinuous square wave. The finite trigonometric sums have value zero at the jump locations, the midpoint of the adjacent target levels.

6.S Chapter 6 supplementary exercises

For real vectors, $u \cdot v = u^T v$ and $\|u\| = \sqrt{u^T u}$. Orthogonal means perpendicular; orthonormal additionally requires unit length. The orthogonal decomposition $x = p + z$, with $p \in W$ and $z \in W^\perp$, has $p = \text{proj}_W x$. It yields the Pythagorean identity $\|x\|^2 = \|p\|^2 + \|z\|^2$. A least-squares solution is an *input* $\widehat{x}$ minimizing $\|b - Ax\|$; the corresponding closest *output* is $A\widehat{x}$. These two vectors can even have different numbers of coordinates.

Exercise 6.S.1: Nineteen true–false statements about orthogonality

(a) **False.** The zero vector has length zero. All lengths are nonnegative; only nonzero vectors have strictly positive length.

(b) **True.** $\| - v\| = | - 1|\|v\| = \|v\|$.

(c) **True.** This is the Euclidean distance definition.

(d) **False.** The rule is $\|rv\| = |r|\|v\|$. A negative r and nonzero v disprove the stated formula.

(e) **False.** The zero vector is orthogonal to every vector, but a set containing zero is dependent. Two *nonzero* orthogonal vectors are independent.

(f) **True.** $x \cdot (u - v) = x \cdot u - x \cdot v = 0 - 0 = 0$.

(g) **True.** Expanding gives $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$; comparison forces $u \cdot v = 0$.

(h) **True.** The analogous expansion with a minus sign gives $\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2$, again forcing zero dot product.

(i) **False.** The projection onto nonzero u is $((y \cdot u)/(u \cdot u))u$, a multiple of u, not generally of y. For $y = (1, 1)$, $u = (1, 0)$ the projection $(1, 0)$ is not a multiple of y.

(j) **True.** A projection onto W belongs to W by its construction; equality with that projection places y in W.

(k) **True.** The set is the null space of the one-row matrix v^T for the fixed vector v, so is a subspace. This includes $v = 0$, when the set is all of $\mathbb{R}^n$.

(l) **False.** Their intersection is $\{0\}$, not empty. If x belongs to both, $x \cdot x = 0$, hence $x = 0$.

(m) **True.** For $i \neq j$, $(c_i v_i) \cdot (c_j v_j) = c_i c_j (v_i \cdot v_j) = 0$. Zero coefficients may introduce zero vectors, but pairwise orthogonality still holds.

(n) **False.** Orthonormal columns give $U^T U = I$. For $U = (1, 0)^T$, $UU^T = \text{diag}(1, 0) \neq I_2$. Squareness is needed to infer the reverse identity.

(o) **False.** Orthogonal columns need not have length one. The matrix $2I$ has orthogonal columns but $U^T U = 4I \neq I$.

(p) **True.** Square orthonormal-column U satisfies $U^T U = I$, hence $U^{-1} = U^T$ and $UU^T = I$. The latter identity says its rows are orthonormal.

(q) **True.** The projection and its residual are perpendicular, so the displayed identity is precisely Pythagoras.

(r) **False.** The minimizing input is $\widehat{x}$; $A\widehat{x}$ is the closest vector in $\operatorname{Col} A$. The inequality is valid, but it describes the output, not the least-squares solution itself.

(s) **False.** The normal equations are $A^TA\widehat{x} = A^Tb$. The inverse formula $\widehat{x} = (A^TA)^{-1}A^Tb$ requires independent columns. For $A = 0$, least-squares solutions exist but A^TA is not invertible.

Exercise 6.S.2: Inductively extend Pythagoras

For $p = 2$, orthonormality makes c_1v_1 perpendicular to c_2v_2; therefore $\|c_1v_1 + c_2v_2\|^2 = c_1^2\|v_1\|^2 + c_2^2\|v_2\|^2 = c_1^2 + c_2^2$. Assume the formula holds for $p-1$ terms. Let $w = \sum_{i=1}^{p-1} c_iv_i$. Since $w \cdot v_p = \sum_{i<p} c_i(v_i \cdot v_p) = 0$, Pythagoras gives

$$\|w + c_pv_p\|^2 = \|w\|^2 + c_p^2\|v_p\|^2 = \sum_{i=1}^{p-1} c_i^2 + c_p^2.$$

This completes induction. For real coefficients $c_i^2 = |c_i|^2$, so the result has exactly the form in the exercise.

Exercise 6.S.3: Bessel's inequality

Let $p = \sum_i (x \cdot v_i)v_i$, the projection of x onto the span of the orthonormal vectors. The residual $z = x - p$ is orthogonal to every v_i, because $z \cdot v_j = x \cdot v_j - (x \cdot v_j) = 0$. Thus $p \perp z$, and Exercise 2 yields

$$\|x\|^2 = \|p\|^2 + \|z\|^2 = \sum_i |x \cdot v_i|^2 + \|z\|^2 \geq \sum_i |x \cdot v_i|^2.$$

Nonnegativity of the residual's squared length is the entire source of the inequality. Equality holds precisely when x lies in the given span.

Exercise 6.S.4: An orthogonal map preserves an orthonormal basis

For each pair i, j,

$$(Uv_i) \cdot (Uv_j) = v_i^TU^TUv_j = v_i^Tv_j = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Thus the n transformed vectors are orthonormal. They are independent: a zero combination, dotted with each transformed vector, forces each coefficient to vanish. An independent set of n vectors in $\mathbb{R}^n$ is a basis, proving the full claim.

Exercise 6.S.5: Preserving all inner products characterizes orthogonality

Apply the given identity to $x = e_i$, $y = e_j$. The vectors Ue_i, Ue_j are columns i, j of U, so their dot product is $e_i \cdot e_j$, equal to 1 for $i = j$ and zero otherwise. Hence the columns of U are orthonormal: $U^TU = I$. Since U is square, this is exactly the condition that U is orthogonal.

Exercise 6.S.6: Real eigenvalues of an orthogonal matrix

Suppose $Uv = \lambda v$ with $v \neq 0$ and real λ. Orthogonal matrices preserve length, giving $\|v\| = \|Uv\| = \|\lambda v\| = |\lambda|\|v\|$. Divide by $\|v\| > 0$ to obtain $|\lambda| = 1$. For a real number this says $\lambda = 1$ or -1. It does not claim every eigenvalue is real: planar rotations can have nonreal eigenvalues of modulus one.

Exercise 6.S.7: A Householder matrix is orthogonal

Put $P = uu^T$ with $u^T u = 1$. Then $P^T = P$ and $P^2 = u(u^T u)u^T = P$. For $Q = I - 2P$, symmetry gives $Q^T = Q$ and multiplication gives $Q^2 = I - 4P + 4P^2 = I$. Consequently $Q^T Q = Q^2 = I$, so Q is orthogonal. This proves length preservation as well as its being its own inverse.

Exercise 6.S.8: A length-preserving linear map is orthogonal

(a) Recover dot products from lengths.

By expansion, $2a \cdot b = \|a+b\|^2 - \|a\|^2 - \|b\|^2$. For $a = T(x)$, $b = T(y)$, linearity and length preservation give

$$2T(x) \cdot T(y) = \|T(x+y)\|^2 - \|T(x)\|^2 - \|T(y)\|^2 = \|x+y\|^2 - \|x\|^2 - \|y\|^2 = 2x \cdot y.$$

In particular perpendicular inputs have perpendicular outputs.

(b) Apply this to the standard basis.

Each $T(e_i)$ has length one, and distinct $T(e_i), T(e_j)$ are perpendicular by part (a). These vectors are the columns of the square standard matrix, so that matrix has orthonormal columns and is orthogonal.

Exercise 6.S.9: Best approximation without orthogonalizing the generators

Set $A = [u\ v]$. The problem is to minimize $\|z - Ac\|$ for $c = (x_1, x_2)^T$. Solve the two normal equations

$$\begin{bmatrix} u \cdot u & u \cdot v \\ u \cdot v & v \cdot v \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} u \cdot z \\ v \cdot z \end{bmatrix}.$$

Their coefficient matrix is invertible because u, v are independent: for nonzero c, $c^T A^T A c = \|Ac\|^2 > 0$. The desired closest vector is $x_1 u + x_2 v$. These equations explicitly account for the nonzero cross term $u \cdot v$; using the separate orthogonal-projection coefficients would be wrong when the generators are not perpendicular.

Exercise 6.S.10: Scale the data, scale the least-squares solution

Independent columns imply $A^T A$ is invertible and the solution is unique. Thus replacing b by cb changes the solution to

$$(A^T A)^{-1} A^T (cb) = c(A^T A)^{-1} A^T b = c\widehat{x}.$$

One can also multiply the original normal equations by c and use uniqueness. The assertion even holds for $c = 0$, though the question only requests nonzero scalars.

Exercise 6.S.11: Fit three parallel inconsistent planes

Write $t = x - 2y + 5z$ and $\mu = (a + b + c)/3$. The squared residual is

$$(t-a)^2 + (t-b)^2 + (t-c)^2 = 3(t-\mu)^2 + (a-\mu)^2 + (b-\mu)^2 + (c-\mu)^2.$$

The equality follows by expanding; the mixed terms cancel because $(a-\mu) + (b-\mu) + (c-\mu) = 0$. The last three terms are constant, so the minimum occurs exactly at $t = \mu$. Every such t is attainable, for instance by taking $x = t, y = z = 0$. Therefore all least-squares solutions, and only those, form the plane

$$\boxed{x - 2y + 5z = (a + b + c)/3}.$$

The least-squares input is nonunique because this equation leaves two free variables, even though the best fitted output is unique.

Exercise 6.S.12: A Rayleigh quotient as a one-variable least-squares fit

Assume the proposed approximate eigenvector v is nonzero. Treat $v\lambda = Av$ as a least-squares system with one unknown. Its normal equation is $(v^T v)\lambda = v^T A v$. Since $v^T v = \|v\|^2 > 0$, its unique solution is

$$\boxed{\widehat{\lambda} = \frac{v^T A v}{v^T v}}.$$

This is the Rayleigh quotient. If $Av = \lambda_0 v$ exactly, the quotient returns λ_0. For $v = 0$ every scalar gives the same zero residual, so there would be no meaningful eigenvalue estimate; nonzeroness is essential.

Exercise 6.S.13: Identify the four fundamental subspaces

(a) If $u \in \operatorname{Nul} A$, each row r_i^T of A satisfies $r_i^T u = 0$. A linear combination of those rows is still perpendicular to u. Thus every member of $\operatorname{Row} A$ is perpendicular to every member of $\operatorname{Nul} A$, proving the stated containment.

(b) If $\operatorname{rank} A = r$, rank–nullity gives $\dim \operatorname{Nul} A = n - r$. A subspace of $\mathbb{R}^n$ and its orthogonal complement have dimensions adding to n, so $\dim(\operatorname{Nul} A)^\perp = r$. The row space also has dimension r. A contained subspace with the same finite dimension must be the whole space, so $\operatorname{Row} A = (\operatorname{Nul} A)^\perp$.

(c) Apply this result to A^T. Since $\operatorname{Row}(A^T) = \operatorname{Col} A$, it gives $\operatorname{Col} A = (\operatorname{Nul} A^T)^\perp$, now inside the ambient space $\mathbb{R}^m$.

Exercise 6.S.14: The consistency alternative in orthogonal-complement language

The system $Ax = b$ is consistent if and only if $b \in \operatorname{Col} A$. By Exercise 13(c), this is equivalent to $b \in (\operatorname{Nul} A^T)^\perp$, which says b is orthogonal to every solution of $A^T y = 0$. This proves both directions of the equivalence. Orthogonality to just one selected null vector would not in general be sufficient.

Exercise 6.S.15: A real triangular Schur factorization gives real eigenvalues

The factorization $A = URU^T$ with orthogonal U makes A similar to R. Similar matrices have the same characteristic polynomial. Since R is upper triangular, $\det(R - tI) = \prod_{i=1}^{n}(r_{ii} - t)$. Its n roots, counted with multiplicity, are the real diagonal entries r_{ii}. Hence A has n real eigenvalues counting multiplicities. The restriction to a real, genuinely triangular R matters; real Schur forms for matrices with nonreal eigenvalues use 2×2 blocks instead.

Exercise 6.S.16: The first step in constructing a real Schur factorization

(a) Since $Ue_1 = u_1$ and $Au_1 = \lambda_1 u_1$,

$$(U^T AU)e_1 = U^T Au_1 = \lambda_1 U^T u_1 = \lambda_1 e_1.$$

This is precisely the assertion about the first column. Here $U^T u_1 = e_1$ follows from the orthonormality of the columns.

(b) The block upper triangular form gives

$$\det(U^T AU - tI) = (\lambda_1 - t)\det(A_1 - tI).$$

Similarity makes this equal to $\det(A - tI)$, whose roots, with multiplicity, are the listed $\lambda_1, \ldots, \lambda_n$. Cancelling the polynomial factor $\lambda_1 - t$ leaves the roots $\lambda_2, \ldots, \lambda_n$ for A_1. This argument also handles repeated values of λ_1: only one copy is removed from the multiplicity list.

Exercise 6.S.17: Check a condition-number bound for perturbed data

Use the Euclidean norm throughout and $\kappa_2(A)$ as the condition number. Exact rational solution of the two systems gives

$$x = (3.94, .49)^T, \qquad \Delta x = (-1.04, 1.51)^T.$$

For example $A^{-1} = \begin{bmatrix} -110 & 310 \\ 160 & -450 \end{bmatrix}$, so multiplying by b and Δb verifies both vectors. The measured ratios and bound are

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .461796, \quad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .000154792, \quad \kappa_2(A) \approx 3362.9997.$$

Their product on the right is about .520567, which exceeds .461796. Thus the asserted inequality holds for this example, despite the large relative solution change.

Exercise 6.S.18: The same perturbation with a different original solution

The same inverse and the new right side give

$$x = (-491.17, 713.15)^T, \qquad \Delta x = (-1.04, 1.51)^T.$$

The perturbation is unchanged because A and Δb are unchanged. However the relative ratios use the new original vector:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .00211738, \qquad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .00211778.$$

Multiplying the latter by 3362.9997 gives about 7.12211, much larger than .00211738. The bound is valid but far from attained. A condition number describes worst-case amplification, not the amplification of every particular perturbation.

Exercise 6.S.19: A small change along a relatively harmless direction

Exact solution of the two printed systems gives

$$x = (-8.642, -1730.054, 2368.716, -844.866)^T,$$

$$\Delta x = (.000030, .000140, -.000155, .000059)^T.$$

These terminating decimals are exact for the printed decimal data. Direct multiplication by the given integer A yields the stated b and Δb, respectively, providing an independent check. Numerically,

$$\kappa_2(A) \approx 23682.9279, \quad \frac{\|\Delta x\|_2}{\|x\|_2} \approx 7.17780 \cdot 10^{-8}, \quad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .000283204.$$

The right-hand bound is approximately 6.70710, far larger than the observed $7.18 \cdot 10^{-8}$. Therefore the inequality holds. The original solution is large, while the particular change in it is tiny.

Exercise 6.S.20: A perturbation close to worst-case amplification

With the new right side and perturbation, exact rational arithmetic gives

$$x = (2.541, 1.460, 1.174, -.101)^T, \qquad \Delta x = (-.002322, -.464848, .636450, -.227007)^T.$$

Multiplication checks $Ax = b$ and $A\Delta x = \Delta b$. The condition number is the same as in Exercise 19, since A is unchanged. The ratios are

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .259665, \qquad \frac{\|\Delta b\|_2}{\|b\|_2} \approx 1.09702 \cdot 10^{-5}.$$

The right-hand bound is .259806, slightly larger than .259665. Thus the inequality holds and is nearly sharp for this choice. All four experiments can be reproduced using exact SymPy solves for x and Δx, then NumPy's `linalg.cond(A)` and Euclidean vector norms. Matching the norm used for the condition number and the relative changes is essential.